

SYSTEM ANALYSIS AND DESIGN OF IFIC BANK

Submitted by

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Roll

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Chapter 1

Recognition of Need

1.1 Introduction

IFIC (International Finance Investment and Commerce) Bank PLC is a banking company incorporated in the People's Republic of Bangladesh with limited liability. It was set up at the instance of the Government in 1976 as a joint venture between the Government of Bangladesh and Sponsors in the private sector to work as a finance company within the country and set up joint venture banks/financial institutions abroad.



Figure 1.1: Current logo of IFIC Bank PLC

In 1983, when the Government allowed banks in the private sector, IFIC Bank was converted into a full-fledged commercial bank. The Government of the People's Republic of Bangladesh holds 32.75% of the share capital of the Bank. Sponsors/Directors having vast experience in the field of trade and commerce own 4.11% of the share capital and the rest is held by Institutions both local & foreign and General Shareholders. The Government is being represented by three nominated Directors on the Board of the Bank and as such, the present combined shareholding of the Sponsors/Directors are 36.86% of the total outstanding share capital of the Bank.

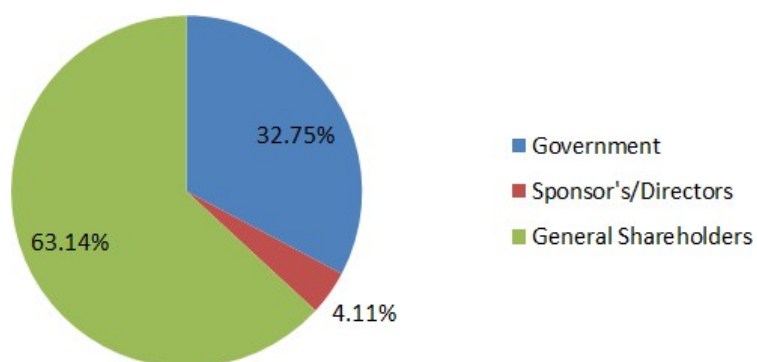


Figure 1.2: Percentage of shares of IFIC Bank PLC

The present capital structure of the bank can be broken down to 40,000 million BDT of authorized capital and 17,859.109 million BDT of Paid-up capital.

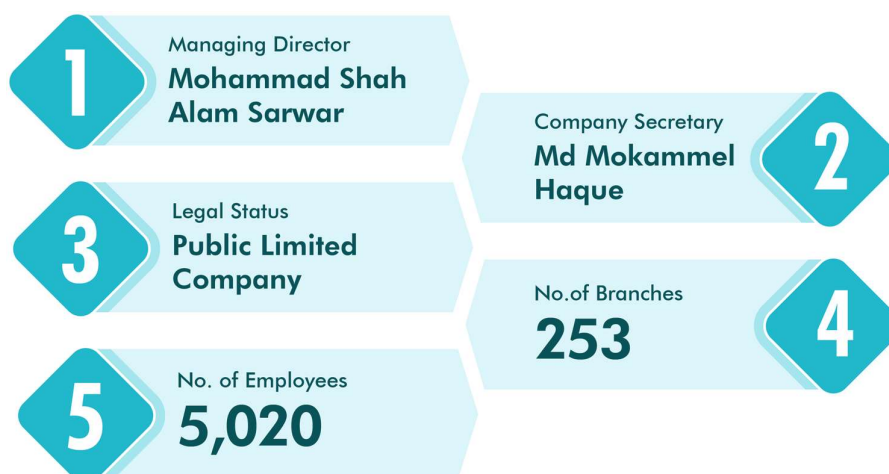


Figure 1.3: The Current Profile of IFIC Bank PLC

The management structure of the Bank is shown below:

The principle activities of the bank revolve around three main departments - General banking department, Credit risk management department and the Foreign exchange department.

IFIC Bank strives to become the preferred financial service provider through innovative, sustainable, and inclusive growth and to deliver the best in class value to all stakeholders. Integrity, fairness, innovation and commitment are regarded as the organisation's core values.

IFIC Bank prioritizes strategies like maintaining adequate capital flow, ensuring effective risk management and diversifying loan portfolios to ensure sound business growth.

From the practical implementation of customer dealing procedures during the whole period of our practical orientation in IFIC Bank PLC, we have learned that being one of the

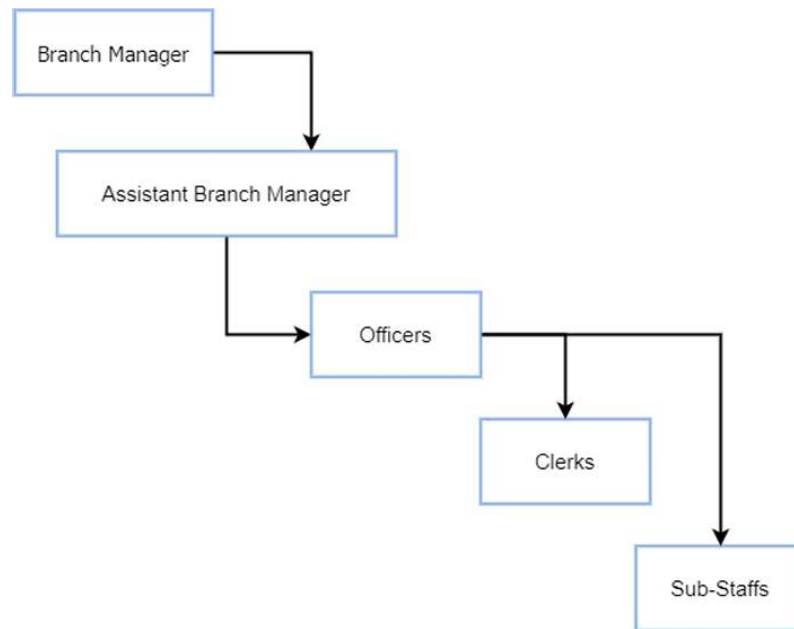


Figure 1.4: Management Hierarchy of IFIC Bank PLC

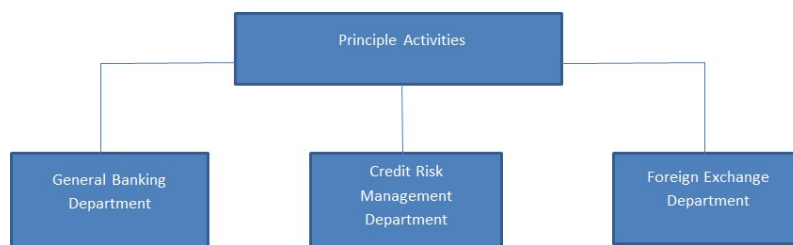


Figure 1.5: Principle activities of IFIC Bank PLC

largest and oldest private commercial banks in Bangladesh, this bank serves people and the government simultaneously.

1.2 Problem Identification

Upon on-site observation, the following problems regarding the banking system of IFIC Bank have been identified:

- **Problem-1: Low interest Rates**

IFIC Bank provides a 2.00% annual interest rate for all savings accounts.

Though a savings account allows for storing money, having some interest from it and taking out the money from the account when needed, holding all money in a savings account probably is not the best option because of the disadvantages of savings accounts.

When interest rates are low, banks typically cannot reduce deposit rates significantly, while the rates they charge on loans also decrease, affecting profitability. Lower interest rates generally encourage borrowing, especially for large purchases like homes or cars. To maintain profitability, some banks may take on higher-risk loans, such as lending to borrowers with weaker credit profiles or offering larger loans. Low interest rates can drive asset prices higher, particularly in real estate or the stock market, as investors search for higher returns. This can result in asset bubbles, which, if they burst, could lead to significant financial instability. Also, In some regions, regulators may require banks to hold higher capital reserves to protect against potential defaults or economic downturns exacerbated by low interest rates. This can limit the amount of money banks can lend and further squeeze profits.

With lower interest rates, customers may find traditional savings accounts or time deposits unattractive due to the minimal returns. This can lead to a reduction in the bank's deposit base, also customer retention.

- **Problem-2: Withdrawal restriction**

This bank's savings accounts are not as flexible when it comes to moving money as checking accounts. Anyone can deposit money in a savings account as many times as he/she wants, but banking regulations restrict to six pre-authorized, automated, or telephone transfers or withdrawals during any given month. But anyone can also do as many transfers as you want to pay a loan at the same bank.

Withdrawal restrictions can lead to a loss of trust between the bank and its customers, as well as increase the chance of damaging the bank's reputation and making it difficult to retain or attract new customers.

If a large number of customers are unable to access their money, it can have broader economic consequences. Reduced spending power can slow down local businesses, leading to job losses and further economic strain in the community. On the other hand, While withdrawal restrictions may temporarily alleviate liquidity pressures, they may not solve the underlying problem if the bank is facing a deeper liquidity crisis. Over time, restrictions can hurt customer retention and reduce the inflow of new deposits, making liquidity issues worse.

Customer retention problems can further worsen the situation, since many customers may not return to the bank for other services.

- **Problem-3: Online and Mobile Banking Issues**

Frequent downtime: IFIC Bank's online and mobile banking services sometimes suffer from outages or slow processing, especially during high-traffic periods, affecting users' ability to perform transactions efficiently.

Limited features: Users may find the IFIC Bank mobile app and online platform especially the UI less user-friendly or lacking in advanced features compared to international banks, which limits the functionality of services like fund transfers, bill payments, and loan applications. For that, it is very hard to know about the current status of someone's own savings account. The existing app does not save any data of transactions. For that, the customers have to re-enter all the information for each transaction such as account number, name, branch name etc which can be tedious. Additionally, the login transition is quite slow. Users can not log in properly if they forget their credentials because the "Forgot Password" option does not work. Though, IFIC Bank provides some messaging facilities to notify about transactions, many banks offer greater interfaces and services through apps for customers.

Technical glitches: Users have occasionally reported bugs in the mobile banking app, such as login issues, delayed transaction notifications, and errors in transaction history. At the local IFIC Bank branch, some common mismanagement issues include:

- **Problem-4: Mismanagement**

Lack of an Information Desk: There is no clear point of contact in IFIC Bank Rajshahi's local branch. Without an information desk, customers often face confusion regarding where to go for basic inquiries or assistance. This lack of direction can lead to customers wasting time while figuring out which service counter to approach. The absence of a dedicated information desk also means that there is no easy way for customers to lodge complaints or get immediate answers to their questions. This can result in frustration, especially for first-time visitors or those unfamiliar with the bank's procedures. In the absence of an information desk, front-line staff may be overburdened with simple inquiries, further slowing down the process for more complex banking needs. There is a lack of basic amenities such as support for first-time users

to navigate the bank's services, such as opening an account, obtaining information on loans, or using digital banking options.

Crowded and Unorganized Waiting Areas: The branch lacks proper management. Without proper management at the branch, customers face chaotic lines or waiting areas, with no staff to guide them or prioritize services based on urgency.

1.3 Conclusion

Through our initial investigation at IFIC Bank PLC, we have tried to define the scope of our analysis and identified several problems within the system of the organization. Addressing such issues, we will carry out further investigation while simultaneously moving along the next stages of the system development life cycle.

Chapter 2

Initial Feasibility Analysis

2.1 Introduction

A feasibility study is a detailed analysis that considers all of the critical aspects of a proposed project to determine the likelihood of its success. Throughout this chapter, we attempt to elaborate on our findings during an initial investigation at IFIC Bank PLC and propose a few suggestions.

2.2 Initial Feasibility study

Our initial investigation has brought forward several drawbacks of the banking system at IFIC Bank PLC. Depending on our initial findings, we attempt to expand our survey into a more detailed study. Considering major factors, we describe the problem and the requirements of a candidate system in the following sections.

2.2.1 Problem-01

2.2.1.1 Problem Statement

IFIC Bank provides lower interest rates compared to other private banks on savings accounts.

2.2.1.2 Summary of Findings

Savings accounts will typically pay more interest than checking accounts, but not as high a rate of return as one can achieve in other types of accounts like. Recently, IFIC Bank has seen a decline in obtaining new customers. According to recent reports, the number of new savings accounts opened at IFIC bank in the 2021-2022 timeline was around 1.2 Lakhs, whereas the number declined to 80,000 thousand in the next year. Lower interest rates can be identified as one of the root causes behind this scenario. The high contrast between the

interest rates of savings accounts and deposit accounts sheds light on the fact that the bank seemingly needs to bring balance to its interest policy.

2.2.1.3 Details of findings

IFIC Bank offers the following interest rates, which impact customer decisions as following:

Savings Accounts: The annual interest rate on savings accounts is 2.00%, which is relatively low compared to other private banks in Bangladesh.

Customer Impact: This lower rate can deter potential customers from opening savings accounts, as they may seek higher returns elsewhere. The unattractive savings rate may slow the bank's ability to attract new depositors, ultimately affecting its growth and customer base.

Deposit Schemes: IFIC Bank provides more competitive interest rates, ranging from 4% to 10% annually, for various deposit schemes.

Customer Appeal: These higher rates offer better returns for long-term depositors, which may attract customers interested in fixed-term investments.

Growth Opportunity: While these deposit schemes are more appealing, the bank's reliance on them to attract customers may not fully offset the impact of low savings account rates. By addressing the disparity between the savings account and deposit scheme rates, IFIC Bank could better balance customer acquisition and long-term business growth.

2.2.1.4 Recommendations

When a bank is operating in a low-interest-rate environment, it faces significant challenges related to profitability, customer retention, and business growth. To cope with these challenges, banks must adopt various strategies to mitigate the effects of low rates. Based on the findings of our initial survey, we recommend the following course of action for the organization-

Diversifying Loan Portfolios: Diversity should be brought in revenue streams to deal with customer losses. Partner with financial technology firms to offer innovative products like digital payments, robo-advisory, or peer-to-peer lending services, which can generate new streams of non-interest revenue.

Introducing Cost-Cutting Techniques: Cost-cutting should be introduced wherever necessary. Automating back-office processes like compliance checks, loan processing, and customer service can lead to significant cost savings.

Attracting Borrowers: Loan-portfolio should be improved focusing on higher-yielding loans. Expand lending into untapped or underserved markets such as small- and medium-sized enterprises (SMEs), digital lending, or green financing (loans for environmentally

sustainable projects)and attracting more borrowers should be encouraged.

Deposit strategy: Prioritizing the increasing demand for low-cost deposits.

2.2.2 Problem-02

2.2.2.1 Problem Statement

The bank imposes inconvenient withdrawal restrictions.

2.2.2.2 Summary of Findings

This bank's savings accounts are not as flexible when it comes to moving money as checking accounts. Anyone can deposit money in a savings account as often as want, but banking regulations restrict to six pre-authorized, automated, or telephone transfers or withdrawals during any given month. But anyone can also do as many transfers as you want to pay a loan at the same bank.

2.2.2.3 Details of Findings

Withdrawal issues can occur when customers face difficulties accessing their funds, and they can have a variety of causes—ranging from operational challenges to liquidity problems. Some of them are discussed below:

Customer Trust: For banks or financial institutions, addressing withdrawal issues is critical to maintaining customer trust, which may falter due to the restrictions.

Liquidity Management: During periods of financial uncertainty or panic, a large number of customers might try to withdraw their funds simultaneously, which can deplete a bank's reserves. Withdrawal restrictions help prevent or slow down such a scenario by limiting outflows and giving the bank time to stabilize.

Restrictions regarding ATMs: Without a MICR check or an ATM card, it is not possible to withdraw money from any branch of IFIC bank other than the specific branch from which the account was created. This creates a substantial issue in user experience leading to inconvenient withdrawal. As a result, the bank fails to make a solid user base among the Middle class and lower class.

2.2.2.4 Recommendations

Dealing with withdrawal restrictions can be complex, but the key solutions should focus on strengthening the bank's liquidity, maintaining customer trust, improving communication, and ensuring regulatory compliance. Below are strategies that banks can adopt to solve or mitigate the problems associated with withdrawal restrictions:

Ensuring Liquidity: Adequate Liquidity must be ensured. Relying on multiple sources of funding, such as wholesale funding, retail deposits, or short-term borrowing, can help banks manage liquidity more effectively and avoid liquidity crunches that lead to withdrawal restrictions.

Monitoring Flow: Cash flow must be monitored. If withdrawal restrictions are mandated or influenced by regulatory requirements, banks should closely coordinate with regulators to ensure compliance while protecting customer interests. Clear communication with regulators also helps avoid legal risks.

Increasing Operational Efficiency: Operational efficiency should be prioritized. If restrictions are necessary, banks must clearly communicate the reasons to customers to avoid panic. Transparency about the bank's financial health and the reasons for restrictions helps maintain trust.

Monitoring Frauds: Fraud and suspicious activities must be prevented.

2.2.3 Problem-03

2.2.3.1 Problem Statement

Online and Mobile Banking Issues

2.2.3.2 Summary of Findings

Key Issue: Frequent outages and slow transaction processing on the bank's online and mobile platforms cause major inconvenience for customers.

Technical Glitches: Users report issues such as login failures, unresponsive apps, and inaccurate transaction histories.

Limited Features: The mobile app lacks some advanced functionalities compared to other banks, such as detailed loan tracking, automated bill payments, and instant notifications for transactions. The UI is very outdated. Customers need to repeatedly enter login details and account numbers which is tedious.

2.2.3.3 Details of Findings

As per our investigation, the following issues have come up regarding online and mobile banking issues:

Frequent Downtime and Slow Processing: Many users have reported issues with the mobile app and online banking platform being down during peak usage times, such as weekends or month-ends. These downtimes delay transactions, frustrate users, and affect overall trust in the system.

Technical Glitches in Mobile App: Users frequently encounter login errors, transaction history inconsistencies, and app crashes. In some cases, customers reported difficulties in

transferring funds or making bill payments due to unresponsive or slow app behaviour.

Lack of Advanced Features: The current online and mobile platforms offer basic functionality but lack advanced services such as automated payments for recurring bills, customizable savings plans, or integrated financial management tools that would enhance the user experience. Additionally, the app lacks real-time transaction alerts and personalized banking offers, making it less competitive compared to other digital banking options available in Bangladesh.

Lack of User-Focused Design: The app's interface is not user-friendly, making it difficult for customers, especially those who are not tech-savvy, to navigate through basic services. The layout is cluttered, and key functions are buried within menus, making the user experience frustrating.

The lack of experienced developers and UI/UX designers is mainly the reason behind the issues.

2.2.3.4 Recommendations

The main issues with the digital banking system mostly revolve around the poor construction and maintenance of the digital platforms hosted by the bank. In order to overcome such problems, we have documented some of our recommendations below:

Improve System Stability and Scalability: Resources to enhance server infrastructure to reduce downtime, particularly during peak usage times should be allocated. Load-balancing techniques should be implemented .

Fix Technical Glitches with Regular Updates: Testing and debugging of the mobile app should be done to resolve existing issues such as login failures, unresponsive interfaces, and delayed transaction processing. A regular update cycle should be implemented to ensure that technical bugs are addressed promptly and the system remains stable.

Expand Mobile and Online Features: Adding more advanced features to the mobile app, such as: Automated bill payments: This allows users to set up recurring payments for utilities or loans. Real-time notifications: This implements instant push notifications for all transactions. Loan and savings management tools: These include options for users to track loan payments, interest accruals, and personalized savings plans.

Revamp User Interface and Experience: Simplify navigation, making key functions like fund transfers, balance checks, and bill payments easily accessible from the home screen. Conducting user experience research can be considered with existing customers to identify pain points and improve the design based on their feedback.

Implementing these recommendations can significantly improve IFIC Bank's online and mobile banking platforms, enhance customer satisfaction, and ensure a more secure and efficient banking experience.

2.2.4 Problem-04

2.2.4.1 Problem Statement

The local IFIC Bank branch suffers from operational mismanagement, most notably the absence of an information desk.

2.2.4.2 Summary of Findings

The lack of an information desk creates disorganization and confusion, causing frustration for customers and inefficiency in branch operations.

Without proper direction, customers are forced to approach multiple counters, leading to overcrowded waiting areas and delays in service. Front-line staff are overwhelmed with basic inquiries, detracting from their ability to efficiently handle more complex banking tasks. Lack of Customer Guidance: First-time visitors, in particular, struggle to understand how to access various services, which affects their overall banking experience.

2.2.4.3 Details of Findings

Our survey has brought forth some of the consequences of the poor mismanagement of the bank, some of which are discussed below:

Reputation Damage: Due to mismanagement, the bank is perceived as unreliable, inefficient, or unsafe and can suffer long-term damage to its brand, which may take years to rebuild.

Poor customer service: Mismanagement can lead to long wait times, unresponsive customer service, and inefficiencies in handling transactions, frustrating customers. In today's competitive banking environment, poor service can quickly lead to customer attrition, as clients move to more reliable and efficient institutions.

Decline in Employee Morale and Productivity: Mismanagement often creates an inefficient work environment where processes are not clearly defined, and employees lack the tools or training necessary to perform their jobs effectively. This can lead to frustration, lower productivity, and a decline in work quality.

Inefficient processes: Poorly managed operations result in inefficiencies, such as slow loan approvals, delayed transaction processing, or cumbersome customer onboarding procedures. Over time, these inefficiencies can cause the bank to lose its competitive edge as more efficient rivals capture market share.

2.2.4.4 Recommendations

Mismanagement and disorganization place a heavy toll on workplace morale. As per our detailed survey, we have put together a set of actions which can be taken by the bank to deal

with such issues. Some of our recommendations are discussed below:

Placing an Information Desk: One of the most important issues to be handled is the absence of an information desk. The desk should be placed directly at the reception of the building, with full-time staff. The staff should be trained on how to greet and interact with customers as well as employees.

Re-organizing office space: Office space should be reorganized to cater for the needs of all kinds of customers simultaneously. The cubicles of the officers should be placed in another room from the transaction processes. Different departments should be placed in different rooms. Enough space to accommodate the waiting lines should be ensured.

Facilities for Customers should be increased: Sitting arrangements for the customers should be improved. Appointments should be introduced. A strict timetable should be maintained to ensure customer satisfaction.

2.3 Conclusion

Throughout this chapter, we have tried to elaborate on the findings of our initial survey and tried to present some recommendations based on our detailed survey.

Chapter 3

Information Gathering & Analysis

3.1 Introduction

Gathering information about existing systems is a key part of analyzing the feasibility of a candidate system. The success of an analyst depends greatly on the proper utilization of the information-gathering tools. Some of the prominent information-gathering tools are interviews, questionnaires and on-site observation. Analysts need the help of these tools to assess the effectiveness of the existing system and design the recommendations for a candidate system.

3.2 Information Gathering

Information can be gathered through available documentation (manual, documents, flowcharts), staff interviews, and on-site observations.

We collected information using two methods:

1. **Online:** Data was gathered from IFIC Bank PLC's official website, which includes details on products, services, and company information. Social media and different search engines also came in handy in the primary information-gathering process. The website comprehensively describes various products, such as savings and deposit accounts, including interest rates and terms.
2. **Offline (Physical):** Additional information was obtained through visits to local branches, where staff provided insights into services and product offerings. Visiting a local branch allowed for face-to-face interaction with bank employees, who can answer specific questions and provide tailored information. Local branches provide firsthand

access to detailed information about the bank's products and services, including current offerings, interest rates, and any promotional deals.

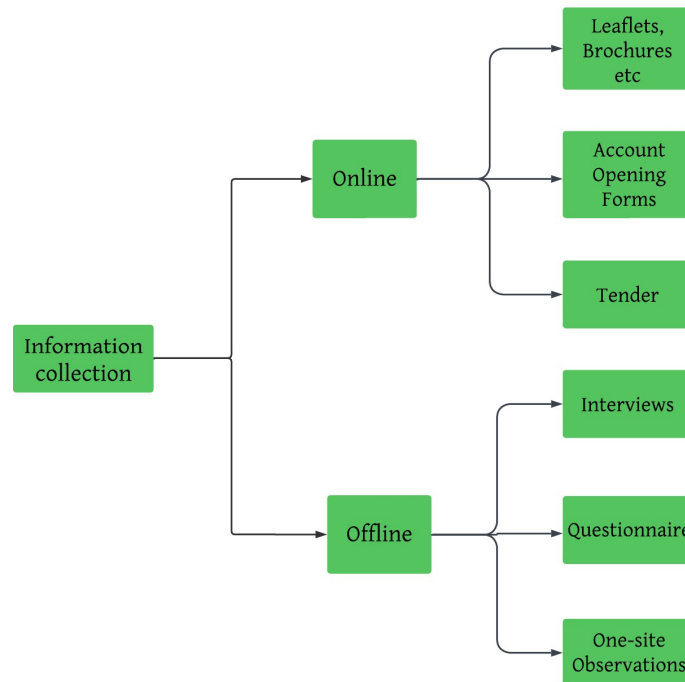


Figure 3.1: Information Gathering Process (Step 1)

3.2.1 Information Gathering Tools

There are several tools for gathering information like Review of literature, procedures and forms, onsite observation etc. We believe information-gathering tools help analysts assess the effectiveness and disadvantages of the current system.

3.2.2 Review of literature, procedures and forms

Procedure manuals and forms are useful sources for the analyst. They can describe the format and functions of the present system. The manual always describes how well the system is designed. Up-to-date manuals save hours of information gathering. To get information about IFIC Bank, we have checked the website, searched the services they provide and checked the reviews of their services.

1. Leaflet

Leaflets are a plan of the marketing plan of IFIC Bank PLC, using which they introduce different bank services to the customer so that the customers can make their own decisions.

If the consumers need any assistance, there are dedicated helpline numbers in the leaflets to help them as well.



**আপনার আস্থা
আমাদের শক্তি**

IFIC

- এক তৃতীয়াংশ (৩৩%) সরকারি মালিকানাধীন সফলতার ৪৬ বছর**
- দেশজুড়ে আমার প্রতিবেশী ১২০০'র বেশি শাখা-উপশাখার সুবিশাল নেটওয়ার্ক**
- আমার একাউন্ট** দেশের সেরা আইএফআইসি আমার একাউন্ট দিচ্ছে এফডিআরের মতো আকর্ষণীয় মুনাফা
- ওয়ার্কস্টপ** প্রতিটি শাখা ও উপশাখায় ওয়ান স্টপ সার্ভিস, একই কাউন্টারে সব ব্যাংকিং সেবা
- আমার কার্ড** দিয়ে দেশের ১৩ হাজারেরও বেশি এটিএম থ্রোক টাকা তোলা একদম ফ্রি
- বেসরকারি ষাতে দেশের সবচেয়ে বেশি হোম লোন দিয়েছে আইএফআইসি আমার বাড়ি**

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www.ificbank.com.bd



Figure 3.2: Leaflet of IFIC Bank PLC

Figure 3.3: Account Opening Form (Page 1) of IFIC Bank PLC

C. Nominee Information											
I/We, nominate the following person(s) to receive/draw the balance held in my/our account after my/our death. (We reserve the right to cancel or change this nomination at any time. (We hereby accord my/our consent that the bank shall not be liable in any manner whatsoever to make payment as per my/our instruction.											
Name of the Nominee											
Address											
Percentage											
Relationship with A/C Holder											
Date of Birth	D	D	M	M	Y	Y	Y	Y			
Identity Document	☐ NID ☐ Passport ☐ Birth Certificate ☐ Driving License ☐ Employee ID ☐ Student ID										
Identity Document Number											
If the nominee remains minor during settlement of the deposit amount after my/our death, (We hereby authorize the following person to receive/draw the amount as per Bank Company Act, 1991 section 103(2).											
Name of Authorized Person	Relationship with Nominee										
Permanent Address											
Identity Document	☐ NID ☐ Passport ☐ Birth Certificate ☐ Driving License ☐ Employee ID ☐ Student ID										
Identity Document Number											
FATCA Declaration	Are you a US resident?	Y	N	Do you hold a US Permanent Resident Green Card?	Y	N					
	Are you a US citizen?	Y	N	Do you have a US residence address?	Y	N					
	Do you have a US correspondence address or a US telephone number?			Y	N						
	US Social Security Number										
Hereby confirm the information provided above is true, accurate & complete. Subject to applicable laws, I hereby give consent to IFIC Bank PLC or any of its affiliates (including branches) (collectively 'The Bank') to share my information with domestic or overseas regulators, or tax authorities or other concerned authorities where necessary to establish my tax liability in any jurisdiction. Where required by domestic or overseas regulator or tax authorities, I consent and agree that the bank may withhold and pay out, from my account(s) such amounts as may be required according to applicable laws, regulations, agreements with regulators or authorities and directives. I agree and undertake to notify the bank within 30 calendar days if there is a change in any information which have been provided to the bank.											
D. Declaration & Signature											
I/We, hereby am/are declaring that, the above information is true. I/We are ready to provide further Information/documents as per requirement of IFIC Bank PLC.											
Signature Admitted/Verified By Bank Official Account No. : Mode of Account Operation : ☐ Singly ☐ Jointly ☐ Either Survivor ☐ Anyone ☐ Others If no. of Account Holders is more than two Branch/Uposhakha Code : Name : EID : PA No. : Signature :	1 st Applicant signature			2 nd Applicant signature			3 rd Applicant signature				
	Please affix here recent, Passport Size Photograph			Please affix here recent, Passport Size Photograph			Please affix here recent, Passport Size Photograph				
	Signature			Signature			Signature				
	Signature			Signature			Signature				
	Name : Customer ID : Mobile No :			Name : Customer ID : Mobile No :			Name : Customer ID : Mobile No :				
E. For Bank (Branch/Uposhakha) Use Only											
Industry Scale ID			WLC Checked	☐ Yes ☐ No	If Yes,	☐ WLC False Positive ☐ WLC Positive					
Account Marketed By	☐ Bank Official ☐ By DSA ☐ N/A		Sector Code								
Name of Account Marketing Official/ DSA			Employee ID/ DSA Code								
Account Opening Officer						Account Approving Officer					
Signature						Signature					
Name : EID : Date :						Name : EID : Date :					

Figure 3.4: Account Opening Form (Page 2) of IFIC Bank PLC

3. Tender

Tenders are public invitations for suppliers to submit a bid to supply specific goods or services. A transparent, efficient tender process enhances a bank's image by demonstrating fairness, transparency, professionalism, and commitment to security and compliance.



IFIC Bank PLC

IFIC Tower, 61 Purana Paltan, GPO Box: 2229, Dhaka 1000. 🌐 www.ificbank.com.bd

INVITATION OF TENDER

IFIC Bank PLC is hereby inviting tender for the following:

Name of Procuring Entity	IFIC Bank PLC			
Invitation For	Proposal for Consultancy, Gap Analysis, Remediation Plan and Rectification of Gaps, Implementation and Certification of PCI PIN for IFIC Bank PLC			
Invitation Ref. No.	IFIC/HO/S&ED/CP/OTM/Tender/PCI-PIN/2023/033			
Procurement Method	Open Tendering Method			

Details of Tender Schedule:

Date of Invitation & Publication	Last Selling Date	Pre-Tender Meeting (If required)	Submission Date and Time	Opening Date and Time
13.11.23	19.11.23	22.11.23 (Optional)	28.11.23 within 2.00 PM	28.11.23 after 3.00 PM

Name & Address for Tender Document:
Tenderer shall submit a request to get Tender Document on official letterhead and collect the Tender Document from Level- 01, IFIC Tower, 61 Purana Paltan, Dhaka.

Eligibility of Tenderer:

- Minimum 03 years continuous experience in the similar line of business.
- At least 05 experience in similar type of work.
- Minimum three technical resources of PCI QSA (Payment Card Industry Qualified Security Assessor) with similar experience in PCI PIN related consultancy/implementation/certification in Bangladesh.
- Tenderer or the QSA partner of the tenderer (Whom the tenderer is representing) must be listed in PCI SSC Council (https://listings.pcisecuritystandards.org/assessors_and_solutions/qpa_assessors).
- Tenderers must have the legal capacity to enter the contract under the applicable law of Bangladesh.
- Tenderers shall not be barred as per law of the land that may subject to legal proceedings of any kind.

Price of Tender Document:
BDT. 3,000.00 (Three Thousand Taka) only (Non-Refundable) to be made in the form of payment order of any scheduled Bank favoring IFIC Bank PLC

Contact Detail of Official Inviting Tender	Name : Mohammad Raisul Islam Email : raisul.islam@ificbankbd.com Phone : 01322-876591	Name : A S A Tousif Ahmed Email : tousif@ificbankbd.com Phone : 01914-868763
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IFIC Bank PLC reserves the right to accept or reject any or all the tenders without assigning any reason whatsoever at any stage of the procurement process.

The Head of Service & Estate Division
IFIC Bank PLC



Figure 3.5: Tender of IFIC Bank PLC



IFIC Bank PLC
IFIC Tower, 61 Purana Paltan, GPO Box: 2229, Dhaka 1000. 📞 www.ificbank.com.bd

Amendment of the tender notice published on 14th September 2023: Vulnerability Analysis, Penetration Testing, Remediation and Performance Testing of New Digital Channel Application for IFIC Bank PLC

Name of Procuring Entity	IFIC Bank PLC
Invitation For	Vulnerability Analysis, Penetration Testing, Remediation and Performance Testing of New Digital Channel Application
Invitation Ref. No.	IFIC/HO/S&ED/CP/Tender/VAPT/2023/026
Procurement Method	Open Tendering Method
The following amendment has been made to the tender document:	
Updated Submission Date and Time	03 October 2023, within 1.00 PM

The Head of Service & Estate Division
IFIC Bank PLC

Figure 3.6: Tender of IFIC Bank PLC

4. Website Structure

Figure 3.7: Landing Page of IFIC Bank PLC with different banking services

5. Mobile Banking

This is our main point of focus. In the app store, many consumers have complained about different issues of the IFIC Aamar Bank. So, we tried to investigate the problem. But, unfortunately, without an existing account, it is not possible to enter into their app. So, we tried to gather information and reviews which are available online and come up with a solution based on what we got. These are some UI of existing mobile applications.

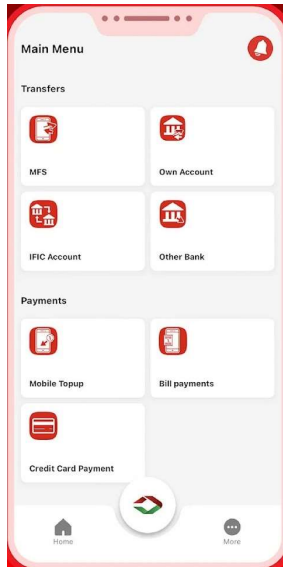


Figure 3.8: Main Menu

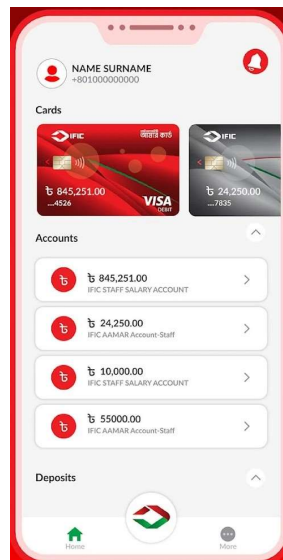


Figure 3.9: Account

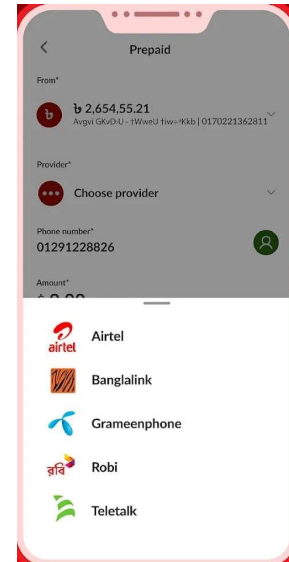


Figure 3.10: Operators

Figure 3.11: UI of IFIC Aamar Bank app

6. Different Schemes Provided by Local IFIC Bank PLC

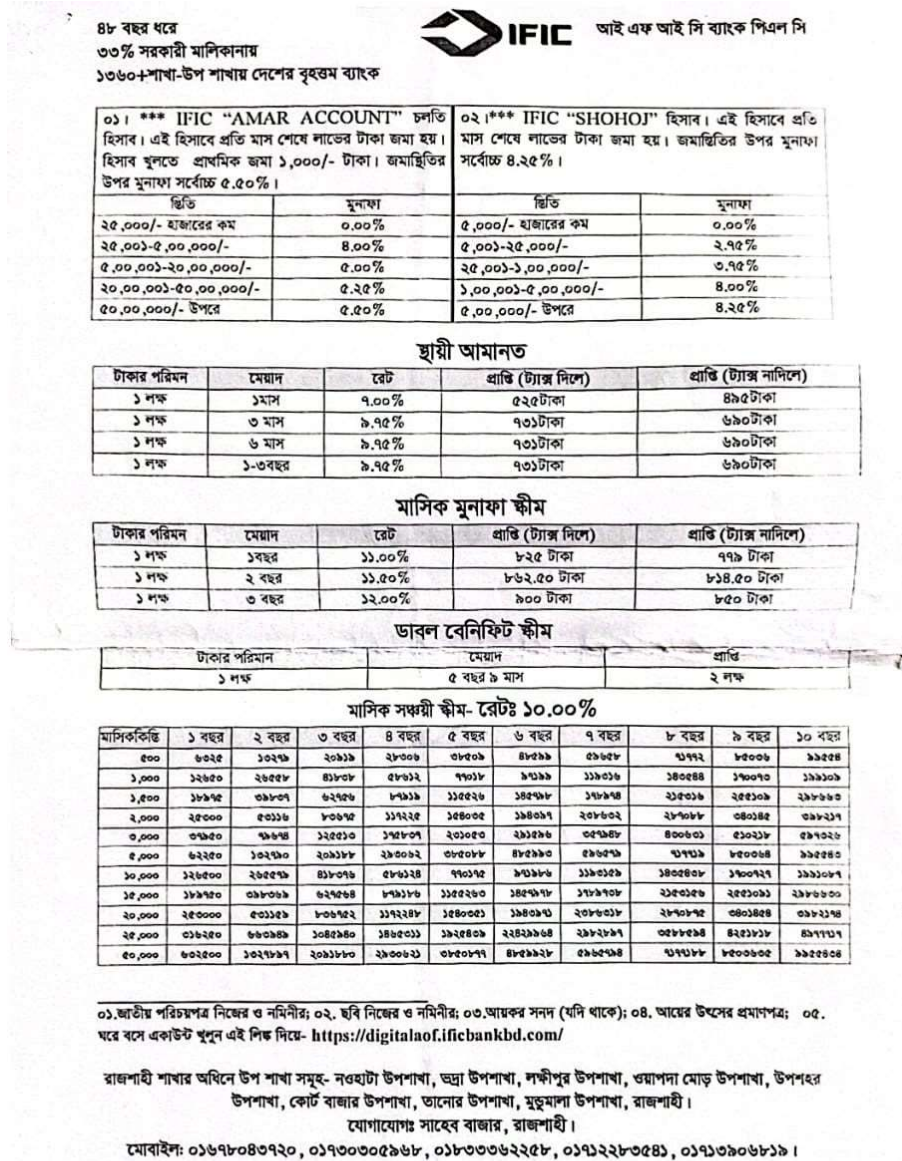


Figure 3.12: Savings Schemes

3.2.3 On-Site Observation

During my visit to the IFIC Bank branch, I made several key observations that highlight areas of strength, as well as areas requiring improvement for enhancing customer experience and operational efficiency. These observations are based on firsthand experience and reflections on the bank's current services, infrastructure, and customer interaction points. Due to Privacy, they did not allow many photographs of the insides to be taken.

Absence of an Information Desk

One of the most striking issues I encountered was the absence of a dedicated information desk or help station. For new or prospective customers, the lack of clear guidance can lead to confusion and frustration.

Delays in Customer Service

The branch staff, although polite, appeared to be overburdened with multiple tasks, resulting in significant delays in customer service.

Comfortable Seating Arrangement

The waiting area had ample seating, with comfortable chairs arranged neatly for customers.

Outdated Account Opening Process

The account opening process remains largely paper-based, as observed from the forms being used.

Cleanliness and Ambience

The overall cleanliness and ambience of the branch were satisfactory, with the premises being well-maintained.

Adequate Lighting

The branch had bright, adequate lighting, which created a welcoming environment and made the space feel open and spacious.

Professional and Friendly Staff Attitude

The staff, despite the rush, maintained a professional and friendly demeanour.

Availability of ATM Services

The branch had functional ATM services available both inside and outside the branch premises.



Figure 3.13: Outside of IFIC Bank PLC, Local Branch



Figure 3.14: Inside of IFIC Bank PLC, Local Branch



Figure 3.15: Inside of IFIC Bank PLC, Local Branch

3.2.4 Interviews

3.2.3.1 Interview with Branch Manager

Question: Thank you for sitting down with me today. I'd like to start by understanding the overall structure of this branch. Could you describe how your bank is organized internally?

Branch Manager: It's my pleasure. At the top level, we have a branch management team, which I lead as the Branch Manager. I oversee operations and ensure that we meet our targets in terms of both customer service and compliance. Under me, we have several key departments: customer service, loan and credit services, account management, and operations, each with its own manager.

Question: How do these departments interact with each other on a daily basis?

Branch Manager: The customer service department is typically the first point of contact for our clients. For example, if a client is interested in a loan, customer service connects them with our loan and credit services team, which handles everything from loan assessments to approvals. We have regular communication between departments to ensure seamless service.

Question: That sounds well-coordinated. Could you explain the decision-making process within the branch? Are most decisions made locally, or do they require approval from the headquarters?

Branch Manager: Day-to-day operational decisions are made at the branch level. However, for bigger decisions, we follow a structured approval process. These decisions often require us to submit proposals or requests to regional management or headquarters.

Question: And how about the technology you use? How does it support the bank's workflow?

Branch Manager: We have a centralized banking system that all departments access for managing customer accounts, transactions, and internal communication. Our digital platforms, including mobile and online banking, are also integrated into the system, allowing for seamless communication between the branch and our digital customers.

Question: Speaking of digital customers, how is the branch adapting to the increasing shift toward online and mobile banking?

Branch Manager: We've adapted by training our staff to handle both digital and in-person interactions efficiently. We also have a dedicated team that focuses on helping customers transition to digital banking services.

Question: How do you ensure smooth communication between the branch and the headquarters or regional office?

Branch Manager: We have regular meetings with regional managers, who provide updates on policies, new products, and compliance changes. Additionally, we use an internal communications platform where important memos, policy updates, and performance reports are shared. If any urgent issue arises, we have direct lines to key departments at the head office, such as risk management or legal, so that we can act quickly.

Question: Let's talk about customer service. What steps do you take to ensure your branch provides excellent customer service?

Branch Manager: First, we focus heavily on training our staff, not just in technical banking procedures but also in soft skills like communication and problem-solving. We also conduct regular customer satisfaction surveys, which help us identify areas for improvement.

Question: How do you handle operational challenges, such as high customer footfall or staff shortages?

Branch Manager: During peak periods, we monitor footfall patterns and adjust our staffing

accordingly. If we anticipate a surge in customers, such as during holidays or the end of the financial year, we bring in additional staff or extend hours temporarily. For staff shortages, we have a cross-training system.

Question: That's a great strategy. Finally, what are some of the key challenges you face as a branch manager, and how do you address them?

Branch Manager: One of the biggest challenges is balancing the need for excellent customer service with strict regulatory compliance. Another challenge is managing evolving customer expectations, especially with the rise of digital banking. We constantly adapt by offering new services and ensuring our employees are up to date with the latest technology.

Question: Thank you so much for your insights. It's been very informative to hear about the inner workings of your branch.

Branch Manager: You're welcome! It was a pleasure speaking with you.

3.2.3.2 Interview with Bank Employee

Question: Thanks for taking the time to talk with me today. Can you tell me a little about your role at the bank?

Bank Employee: Of course! I'm a Senior Customer Service Representative at our branch. My main responsibilities include assisting customers with their banking needs—whether that's opening accounts, resolving issues, or helping them understand our products like loans, credit cards, or online banking.

Question: That sounds like a busy role. What does a typical day look like for you?

Bank Employee: It definitely keeps me on my toes! Every day starts with reviewing emails and any updates from the branch manager or headquarters. We often have team huddles in the morning to go over our goals for the day. Then, I'm on the floor helping customers—most of whom need help with routine transactions like deposits, withdrawals, or questions about their accounts. I also spend time setting up new accounts and handling customer inquiries over the phone or via email.

Question: So, a lot of customer interaction. What's the most rewarding part of your job?

Bank Employee: I enjoy building relationships with our regular customers too. Many of them come back to me specifically because they trust me to get things done right.

Question: What are some of the challenges you face in your role?

Bank Employee: A big challenge is managing high customer expectations. Another challenge is keeping up with all the new products and changes in the banking sector. Technology is changing fast, and we constantly have to adapt to new systems, processes, and tools.

Question: Speaking of technology, how has digital banking impacted your work?

Bank Employee: A lot of customers now handle their transactions online, so foot traffic in the branch has decreased for routine things like checking balances or transferring money. But at the same time, we've seen an increase in people needing help with things like setting up online accounts or troubleshooting issues with mobile banking.

Question: That's an interesting shift. How do you stay motivated with such a dynamic workload?

Bank Employee: I think what keeps me motivated is the variety in my job and the team I work with. Also, my colleagues are great. W Plus, the bank invests in training us regularly, so I feel like I'm constantly developing new skills.

Question: You mentioned training—what kind of training do you receive at the bank?

Bank Employee: We go through a mix of technical and soft skills training. More recently, we've had sessions focused on digital literacy, where we learn how to navigate our online platforms so we can assist customers better.

Question: It sounds like you deal with a lot of sensitive information. How do you manage security and privacy concerns?

Bank Employee: Absolutely. We follow strict protocols to protect customer information. The bank has invested heavily in cybersecurity measures, and we are trained on best practices to avoid data breaches or phishing attempts.

Question: With all the interactions you have, what's the most important skill for someone in your role?

Bank Employee: I'd say patience and communication. Customers need to feel heard and respected, and that's what builds long-term relationships.

Question: That makes sense. What's the biggest change you've seen in banking during your time here?

Bank Employee: Definitely the shift towards digital and self-service banking. But the interesting thing is, that even though digital banking is more convenient, people still come in for big decisions—like taking out a loan, planning for retirement, or discussing investment options. There's still a demand for face-to-face interaction when it comes to more complex or important financial matters.

Question: It's fascinating how personal banking still remains, even in a digital world. Any advice for someone considering a career in banking?

Bank Employee: Definitely! It's not just about numbers—it's about solving problems, building relationships, and making a real impact on people's financial lives. My advice would be to stay curious, be adaptable, and always be willing to learn.

Question: Great advice. Thank you so much for sharing your experience with me today!

Bank Employee: You're welcome! It was a pleasure speaking with you.

3.2.5 Questionnaire

3.2.4.1 Information Collection form(From Employee)

Please answer each Question by checking one:

1. What kind of services among the following do you mostly provide? (Mark all that applies)

- a. Deposit Accounts
- b. Loans and Credit Services
- c. Payment and Transfer Services
- d. Foreign Exchange and International Services
- e. Insurance Products
- f. Merchant Services
- g. Others, please specify-

2. What are the main goals of your organization? (Mark all that applies)

- a. Providing Quality Services
- b. Making Profit

- c. Bringing impactful change to the Banking industry
- d. Revolutionizing Digital Banking
- e. Others, please specify-

3. Do you think the Bank needs to bring changes in its policies regarding customer service, interest rates and management?

- a. Yes
- b. No
- c. Undecided

4. Do you think the Online and Digital Banking services provided by the Bank, need to be improved?

- a. Yes
- b. No
- c. Undecided

5. Do you think the Bank has been successful in fulfilling its organizational goals and targets, in the last 5 years?

- a. Yes
- b. No
- c. Undecided

6. On a scale of 1 to 5, how much satisfied are you with your job?

- a. 1/5
- b. 2/5
- c. 3/5
- d. 4/5
- e. 5/5

7. On a scale of 1 to 5, how much would you rate your current organization as a Commercial Bank?

- a. 1/5
- b. 2/5
- c. 3/5
- d. 4/5
- e. 5/5

8. Have you ever thought about switching jobs?

- a. Yes

b. No

9. What are the main challenges of working in this bank?

- a. Bad working conditions
- b. Pays low
- c. Unrealistic demands
- d. Employee-Management relationship
- e. Others, please specify

10. On a scale of 1 to 5, how much would you rate the management of this bank?

- a. 1/5
- b. 2/5
- c. 3/5
- d. 4/5
- e. 5/5

11. How would you describe your experience at this bank?

- a. Very Good
- b. Good
- c. Average
- d. Bad
- e. Very Bad

12. Are you planning on staying on this bank for at least the next 5 years?

- a. Yes
- b. No
- c. Undecided

13. What do kind of changes would you like to see in the organization? please describe.

14. What do you think are the main challenges faced by your organization right now? Answer:

15. Are there any additional comments you would like to make about your position or the organization as a whole? Answer:

3.2.4.2 Information Collection Form(From Customers)

1. What kind of services among the following do you receive from the Bank? (Mark all that applies)

- a. Deposit Accounts
- b. Loan and Credit Services
- c. Payment and Transfer Services
- d. Merchant Banking
- e. Others, please specify-

2. Have you ever taken services from another Bank?

- a. Yes
- b. No

3. What made you choose this Bank?(Mark all that applies)

- a. Better Service
- b. More Benefits
- c. Others, please specify-

4. Do you use the Bank's online and digital services?

- a. Yes
- b. No

5. On a scale of 1 to 5, how much would you rate your experience at this bank?

- a. 1/5
- b. 2/5
- c. 3/5
- d. 4/5
- e. 5/5

6. Do you have thoughts about switching banks?

- a. Yes
- b. No
- c. Undecided

7. Would you recommend your friends or family members to take services from this bank?

- a. Yes
- b. No

8. What are the main challenges you face in this bank?

- a. Bad behaviour
- b. Low profit
- c. Mismanagement
- e. Others, please specify

9. On a scale of 1 to 5, how much would you rate the management of this bank?

- a. 1/5
- b. 2/5
- c. 3/5
- d. 4/5
- e. 5/5

10. Are you planning on staying on this bank for at least the next 5 years?

- a. Yes
- b. No
- c. Undecided

11. How would you describe your experience at this bank?

- a. Very Good
- b. Good
- c. Average
- d. Bad
- e. Very Bad

12. What do kind of changes would you like to see in the organization? Please describe.

13. What do you think are the main challenges faced by the organization right now? Answer:

14. Are there any changes you want to see in this organization? If yes, then what kind of changes do you expect?

Answer:

15. Do you have any additional comments about the Bank?

Answer:

3.3 Analysis of Questionnaire

3.3.1 Information Collection Form (From Employees)

Please answer each Question by checking one:

1. What kind of services do you mostly provide among the following?

- a. Deposit Accounts
- b. Loans and Credit Services
- c. Payment and Transfer Services
- d. Foreign Exchange and International Services
- e. Insurance Products

Graph:

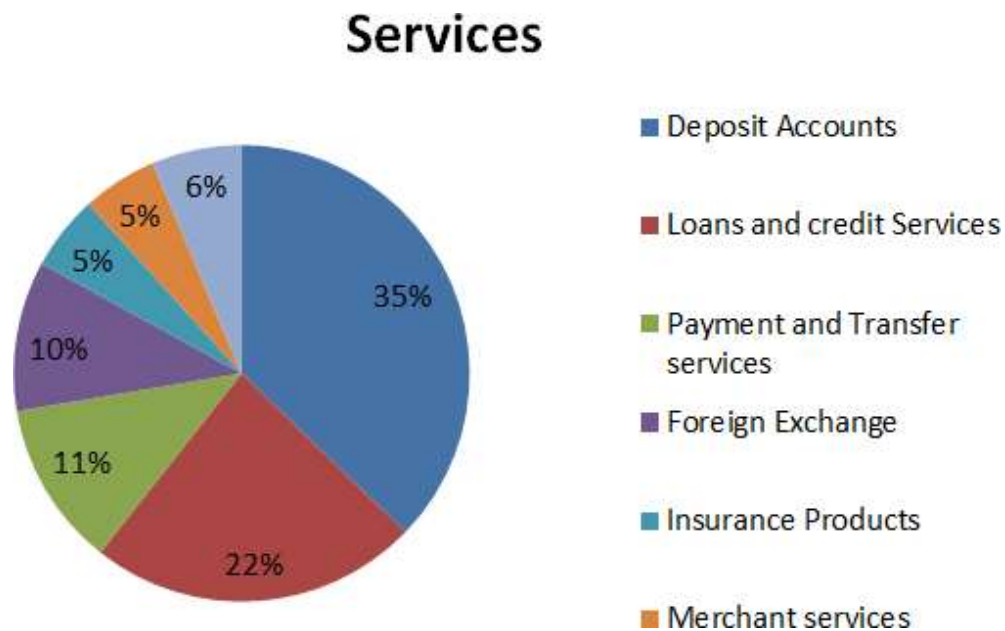


Figure 3.16: Graph of most popular services

Conclusion: Deposit Accounts are the most popular service provided by the bank, closely followed by loans and credit services. The other services closely compete with each other in terms of popularity.

2. What are the main goals of your organization? (Mark all that applies)

- a. Providing Quality Services
- b. Making Profit
- c. Bringing impactful change to the Banking industry
- d. Revolutionizing Digital Banking
- e. Others, please specify-

Graph:



Figure 3.17: Graph of priority of organizational goals

Conclusion: The most prominent organizational goal according to employees is providing quality services to customers. Next comes making a profit, followed by bringing impactful changes and revolutionizing the banking sector.

3. Do you think the Bank needs to bring changes in its policies regarding customer service, interest rates and management?

- a. Yes
- b. No
- c. Undecided

Graph:

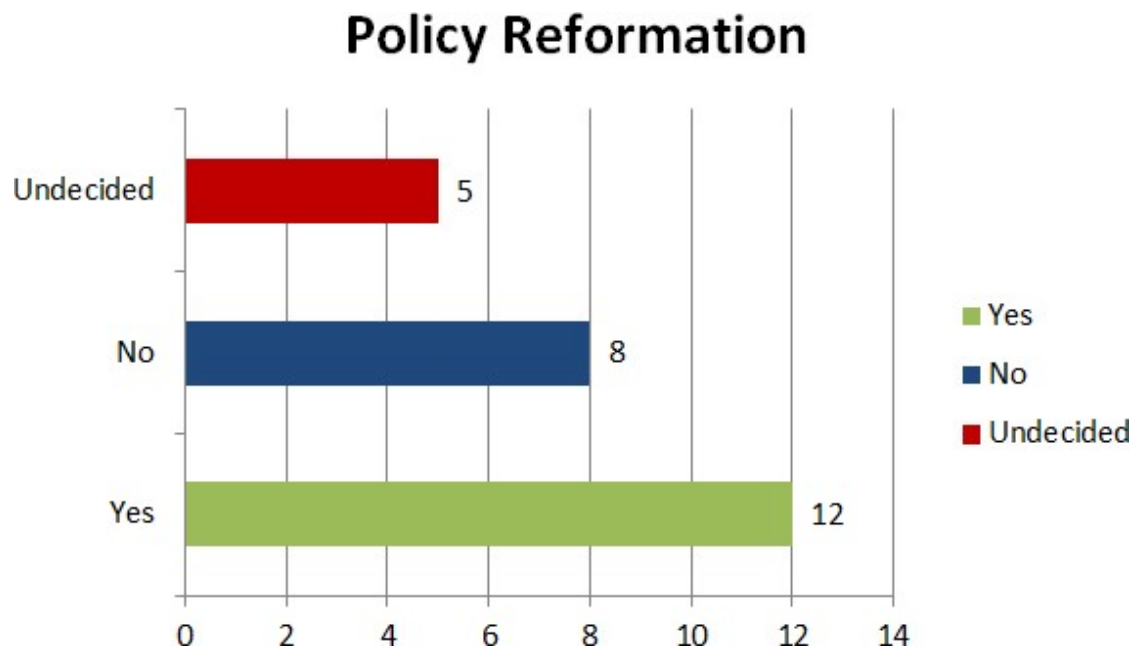


Figure 3.18: Graph of Employee Opinion on Policy Reformation

Conclusion: About 12 people have said yes to policy reformation, on another hand 8 people have voted for no and 5 have been indecisive. The graph highlights the fact that most employees want policy reform. While a significant amount of employees are against reformation, since the majority wants reformation, the bank should keep an eye on that.

4. Do you think the Online and Digital Banking services provided by the Bank, need to be improved?

- a. Yes
- b. No
- c. Undecided

Graph:

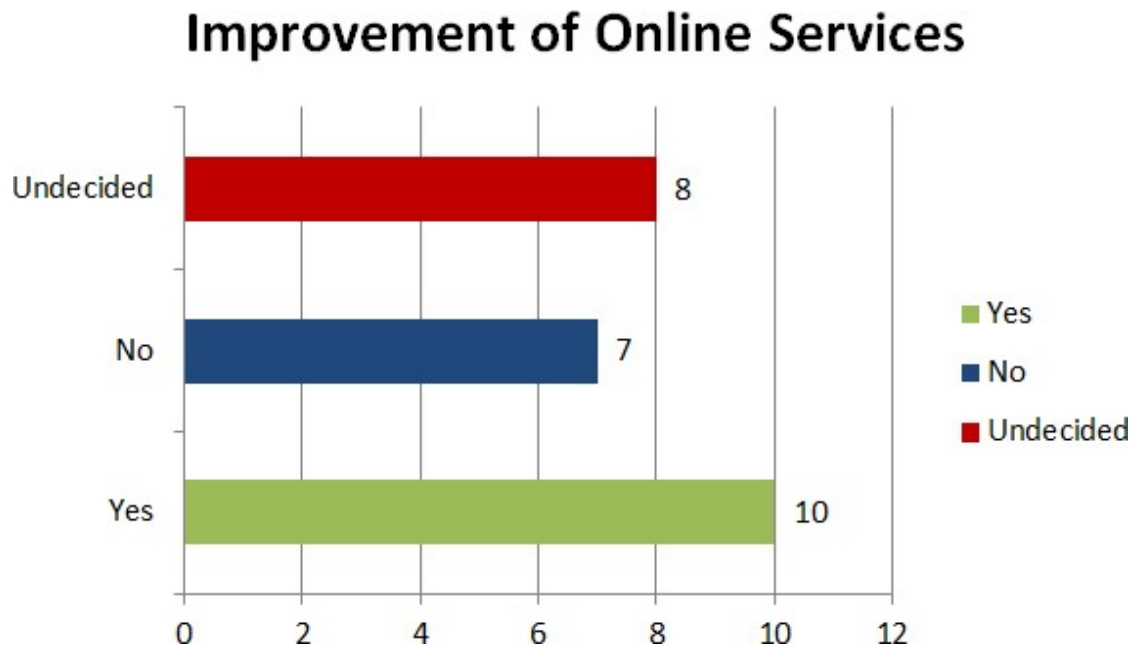


Figure 3.19: Graph of Employees opinion on improvement of online services

Conclusion: About 10 people have said yes to the improvement of online services, on another hand 7 people have voted for no and 8 have been indecisive. The graph highlights the fact that most of the employee wants improvement. Since the majority of the employees want to have improvement on online/digital banking services, it is safe to say the bank should go ahead on that path.

6. On a scale of 1 to 5, how much satisfied are you with your job?

- a. 1/5
- b. 2/5
- c. 3/5
- d. 4/5
- e. 5/5

Graph:



Figure 3.20: Graph of Employees' Job Satisfaction

Conclusion: Most of the employees have neutral opinions about their job experience, hence it can be said that the majority of employees are not completely satisfied with their job. The reasons behind the majority of employees being unsatisfied on their jobs could include lower salary, lack of additional benefits, bad working environment, bad relationship with superiors etc.

7. On a scale of 1 to 5, how much would you rate your current organization as a Commercial Bank?

- a. 1/5
- b. 2/5
- c. 3/5
- d. 4/5
- e. 5/5

Graph:

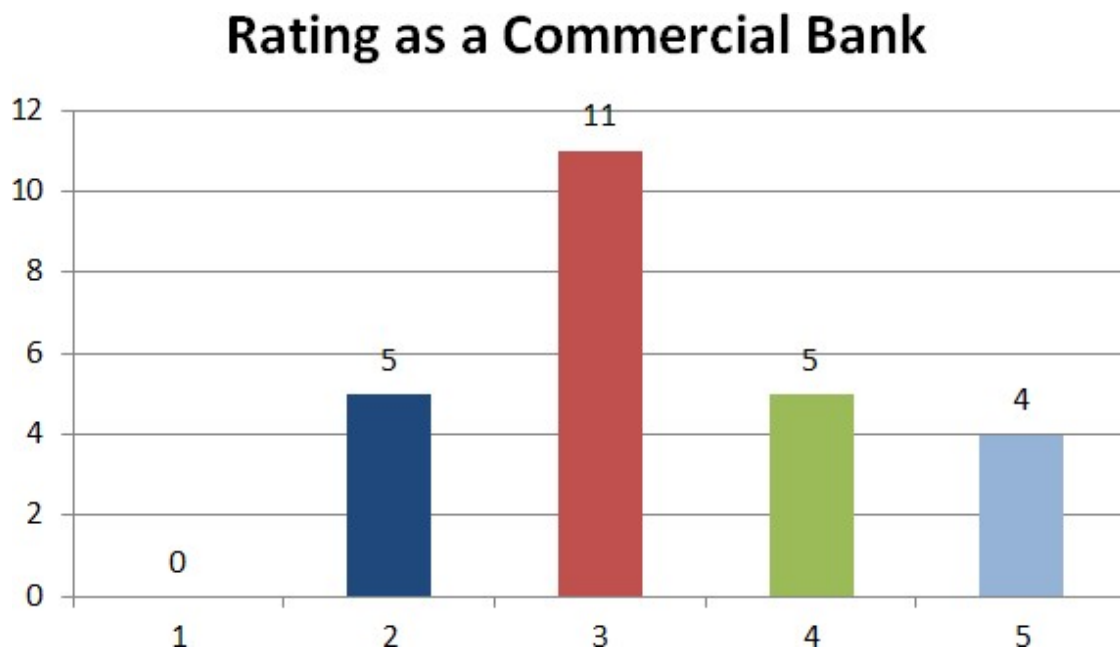


Figure 3.21: Graph of Employees' Opinion on current organization

Conclusion: Most of the employees have neutral opinions about the current organization. In general, most of them believe that the organization is mediocre. Most of the employees are neutral about their job satisfaction. The image of bank in majority of the employees' eyes could be seen through this graph. As the bank undoubtedly suffers from image issue, more attention should be given to image building and branding.

3.3.2 Information Collection Form(From Customers)

Please answer each Question by checking one:

1. What kind of services among the following do you receive from the Bank?

- a. Deposit Accounts
- b. Loan and Credit Services
- c. Payment and Transfer Services
- d. Merchant Banking
- e. Others, please specify- **Graph:**

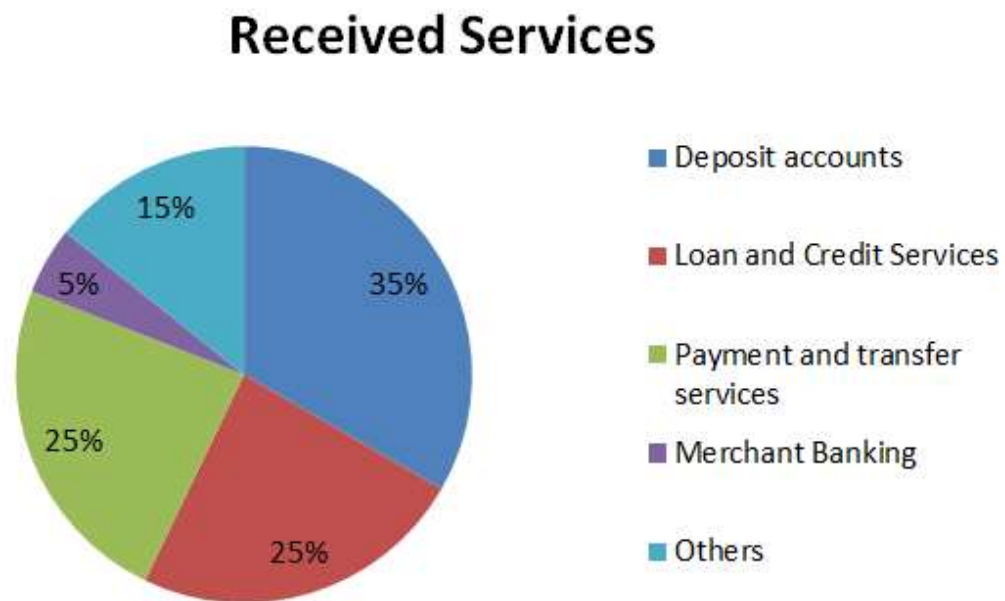


Figure 3.22: Graph of most popular services

Conclusion: Most of the customers are mostly involved with the bank through deposit accounts. Other services like loan and credit services, payment and transfer services etc are relatively less popular among customers.

3. What made you choose this Bank? (Mark all that applies)

- a. Better Service
- b. More Benefits
- c. Others, please specify-

Graph:

Reasons for choosing this Bank

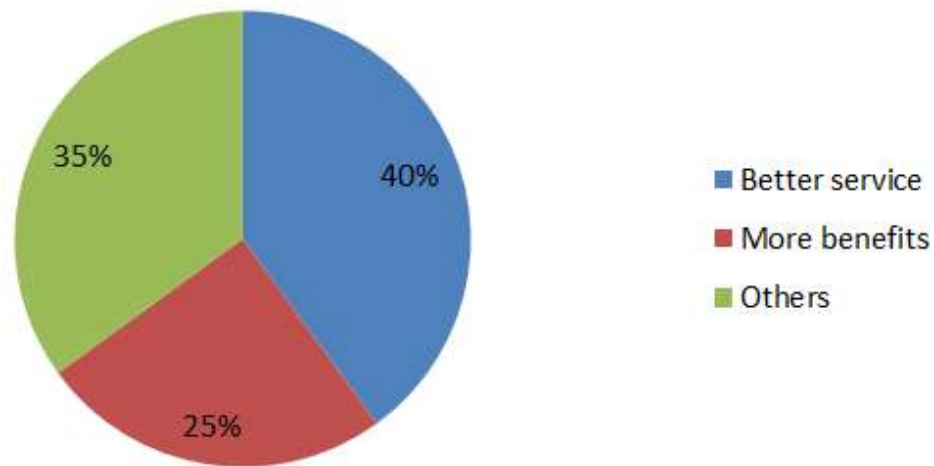


Figure 3.23: Graph of Bank benefits

Conclusion: Customers are mostly attracted to the bank because of its better service quality. While a huge number of customers are also attracted to the bank because of increased benefits and other things, it can be said that to the customer's eyes, the banking services provided by the bank are satisfactory.

4. Do you use the Bank's online and digital services?

a. Yes

b. No

Graph:

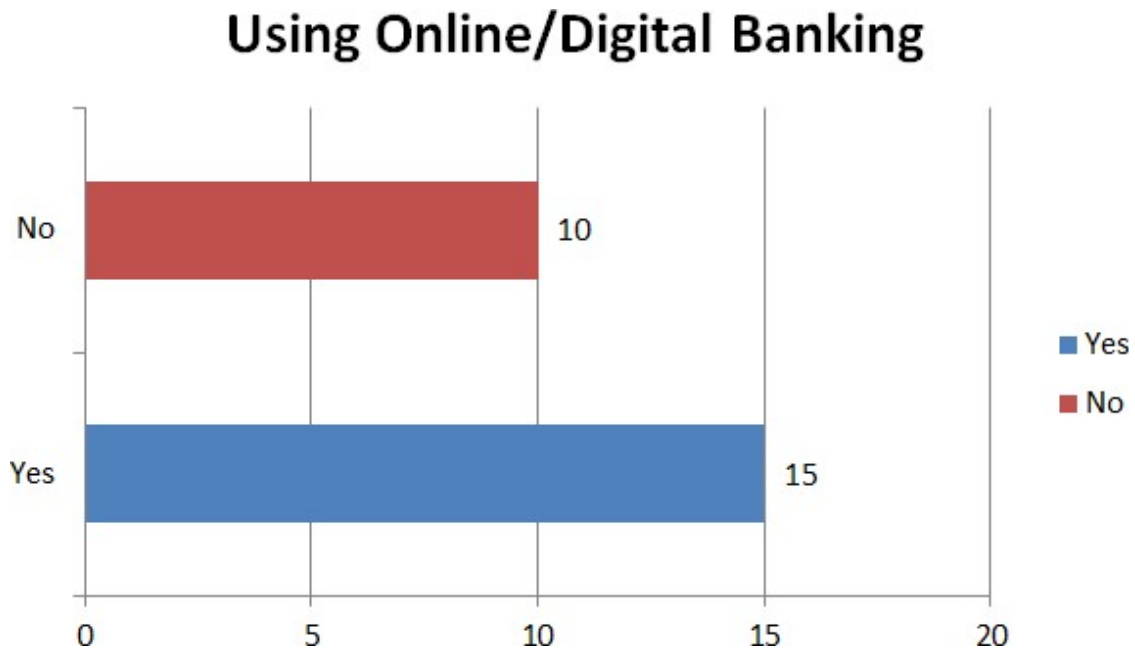


Figure 3.24: Graph of Customers' Digital/Online Service Usage

Conclusion: Most of the customers use the bank's online/digital platforms. Even though a large number of customers do not get involved with the online. digital services provided by the bank, it can be safe to say, that the online/digital platform of the bank is quite popular to the customers.

5. On a scale of 1 to 5, how much would you rate your experience at this bank?

- a. 1/5
- b. 2/5
- c. 3/5
- d. 4/5
- e. 5/5

Graph:



Figure 3.25: Graph of Customer Experience

Conclusion: Most of the customers have average experience with the bank. Although, some of the customers have both relatively good and bad experiences with the bank, the average experience of majority of customers is an alarming concern for the bank authority.

6. Do you have thoughts about switching banks?

- a. Yes
- b. No
- c. Undecided

Graph:

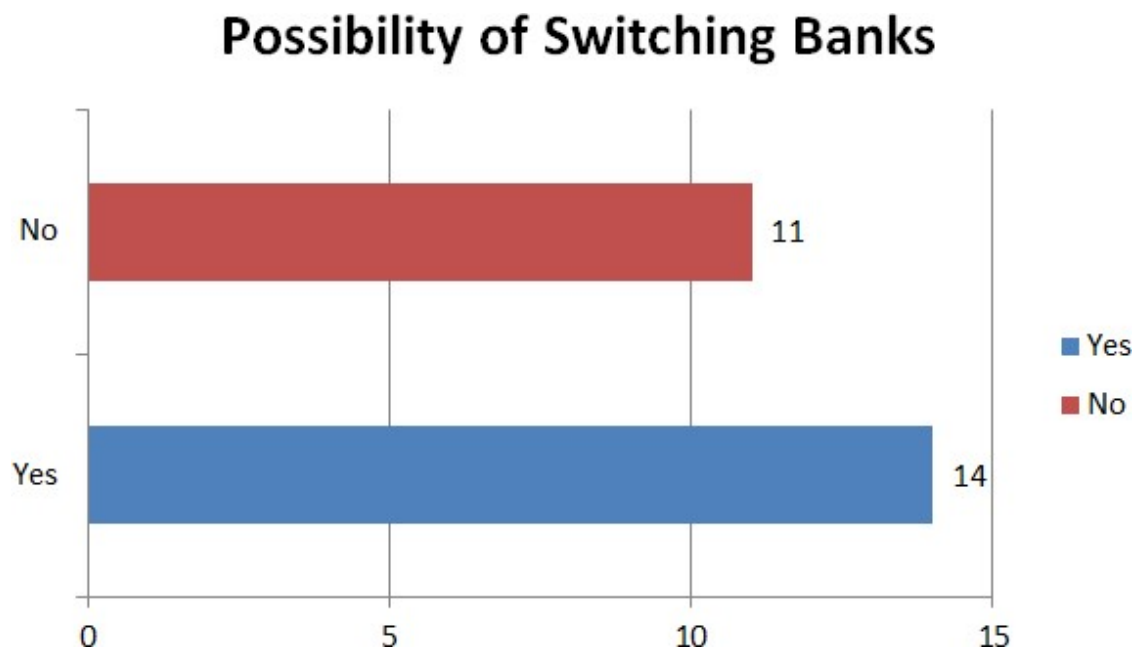


Figure 3.26: Graph of Customers' Probable Switching of Banks

Conclusion: Most of the customers wishes to change banks. The reason behind this could involve factors like low interest rates, bad customer services, inconvenient online platforms, management problems etc. As alarming as this proposition is, the authority definitely should be concerned about this fact, while redesigning their policies.

7. Would you recommend your friends or family members to take services from this bank?

a. Yes

b. No

Graph:

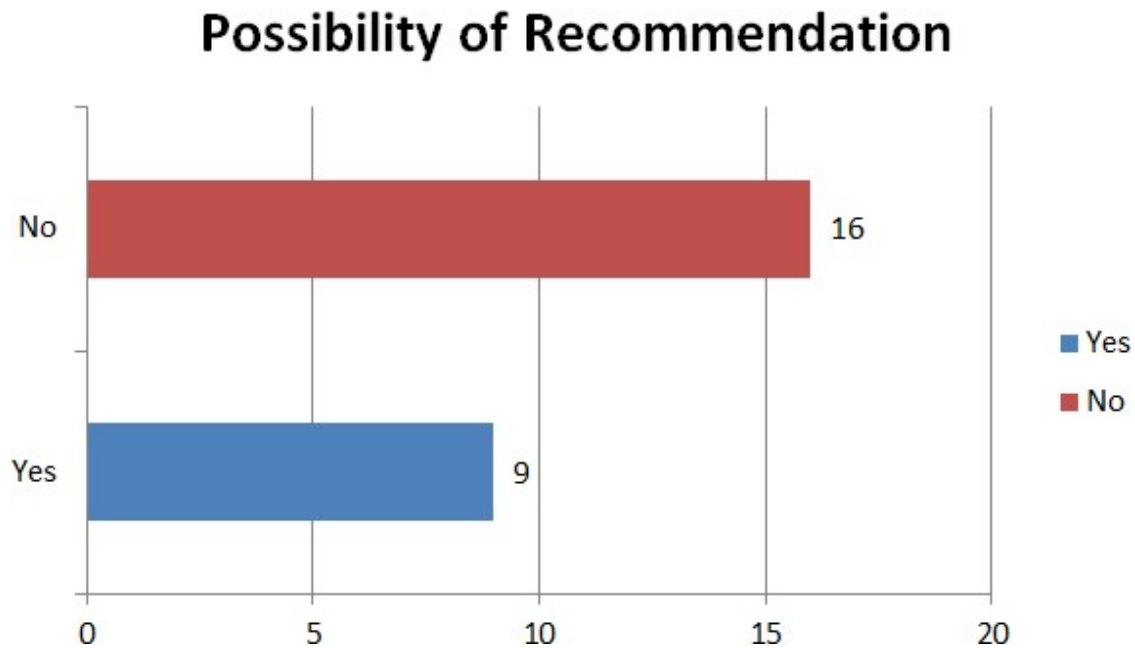


Figure 3.27: Graph of Customers' Probable Recommendation

Conclusion: Most of the customers do not wish to recommend the bank's services to their friends or close ones. The reason behind this could involve factors like low interest rates, bad customer service, inconvenient online platforms, management problems etc. As alarming as this proposition is, the authority definitely should be concerned about this fact, while redesigning their policies.

3.4 DFD of Existing System

3.4.1 DFD of Knowing the Current Account Status of a Customer

This diagram illustrates a flowchart representing a typical banking process for managing account status and related data collection. Here's a breakdown of the key stages:

1. Customer: The process starts with the customer. This represents the user initiating the interaction with the system.

2. Check current status: After the customer enters the system, the first action is to check the customer's current status. This could involve retrieving the customer's account details or assessing their financial situation before proceeding further.

3. Give account info & money: Once the status has been checked, the next step is for the system to provide account information and financial details (e.g., balances, transactions). This interaction might involve both account data and monetary transactions or updates related to the customer.

4. Verify Data: After providing the account information, the next step involves data verification. This step ensures that the data provided (likely from the customer and the system) is correct and consistent. It might involve checking the accuracy of the account details, transactions, or user inputs.

5. Collect Data: Following the data verification, the system proceeds to collect the necessary data. This could include transaction records, user inputs, or other relevant information that the system needs to store or process.

6. Database: Finally, the collected data is stored in a database. This step ensures that all relevant information is securely saved for future use, either for reporting, audit trails, or further customer interactions.

Throughout the flow, arrows indicate the direction of data and process flow between each stage, showing the progression from the initial customer interaction to the final data storage in the database. This flowchart visually organizes the banking process in a step-by-step manner, illustrating the movement of information from the customer to the bank and back.

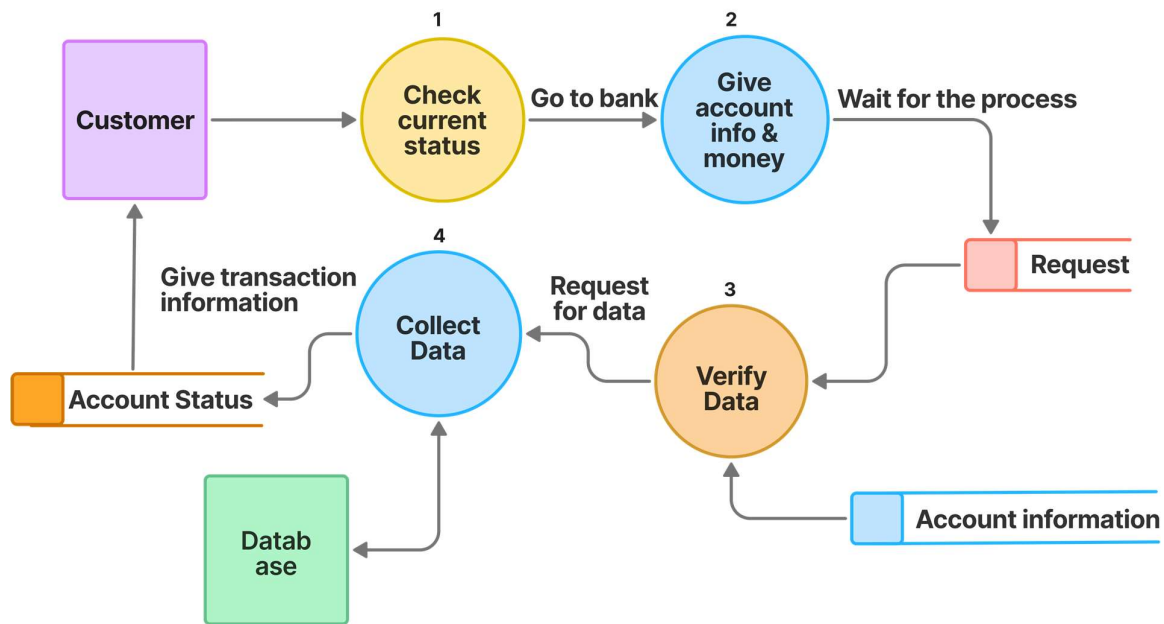


Figure 3.28: DFD For Knowing Current Account of Status

3.4.2 DFD of Money Deposit to the Bank

This diagram represents a basic flow of money deposits and account information updates in a bank system. Here's a description of the process shown:

1. Customer: The process starts with the customer who initiates a transaction. The customer provides transaction information and interacts with the system.

2. Deposit money: The first action after the customer initiates the process is to deposit money. This step suggests that the customer goes to the bank or interacts with the system to submit money for the transaction.

3. Give account info & money: After the deposit, the system gives the customer's account information and the relevant financial details (possibly the deposited amount). This also indicates that the system provides account details as part of the transaction process.

4. Wait for the process: At this stage, the system waits for the process to complete. The request has been submitted, and the system processes the transaction or checks the required data.

5. Verify Data: The system then moves to verify the data. This verification checks that the details provided (such as account information or transaction data) are correct and valid.

6. Request: A request for data or action is initiated, likely as part of the verification or transaction handling. The request flows back to previous steps to ensure the process continues correctly.

7. Update account info: Once the data has been verified, the system updates the ac-

count information. This could mean updating balances, reflecting the deposit, or modifying transaction records.

8. Transaction Information: Transaction information is generated and recorded. This may refer to the actual details of the customer's transaction, which are passed along and stored as part of the overall process.

9. Database: The final step involves storing all relevant information in a database. This ensures that the transaction and updated account data are securely saved for future reference.

Throughout the diagram, arrows indicate the flow of data between the customer, various transaction steps, and the database. Some external interactions such as "Give transaction information" and "Request" suggest the involvement of different systems or components interacting with the main process.

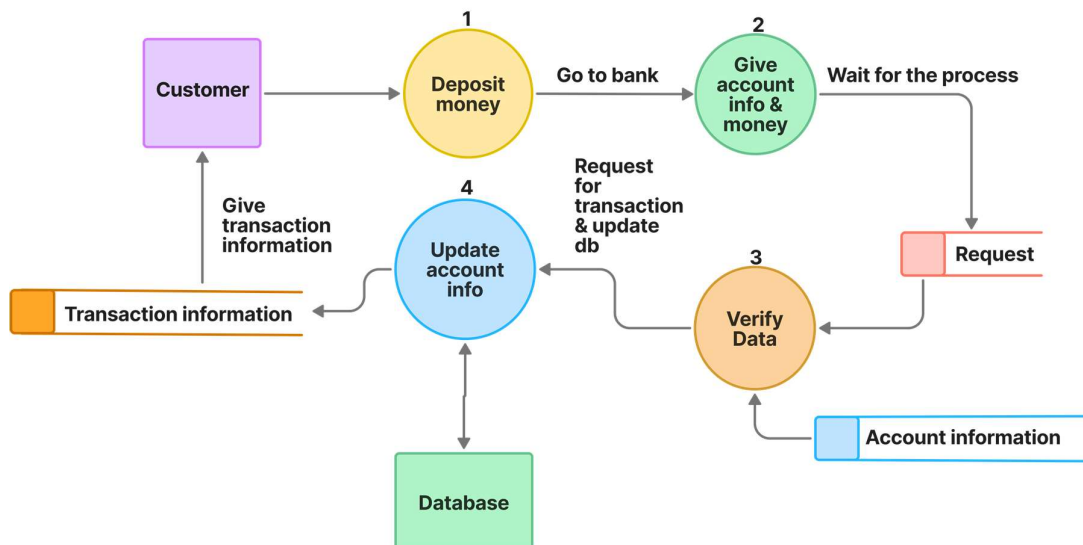


Figure 3.29: DFD For Money Deposit

3.4.3 DFD of Withdrawing Money from the Bank

The diagram outlines the money withdrawal process and highlights several key steps in how a customer interacts with the system, including interactions with bank systems and databases. Here's a detailed breakdown:

1. Withdraw Money: This is the initial action taken by the customer when they want to withdraw money from their account. The customer starts by selecting either of two available paths to process the withdrawal.

2. Two Withdrawal Paths: The diagram shows two distinct methods by which the customer can proceed:

2(a) Give Cheque: The customer may choose to visit the bank and withdraw funds by submitting a cheque.

2(b) Give Account Info: Alternatively, the customer can visit the nearest ATM booth to withdraw funds. At the ATM, the customer would need to provide account information, such as a PIN or card details. Both paths lead to the next phase where a request is sent for further processing.

3. Request Handling: After choosing either the cheque or ATM route, a request is made for withdrawal processing. This involves communication between the customer's interaction and the back-end system, which checks whether the request can be fulfilled based on the account's balance and status.

4. Verify Data: The system moves on to a verification phase, where the provided account information, withdrawal amount, and other transaction-related data are validated.

This step ensures that the customer's request is legitimate and that they have sufficient funds available for withdrawal. The account information is retrieved from the bank's database for cross-verification.

5. Update Account Info: Once the verification is successful, the system proceeds to update the account information. This involves: Deducting the withdrawn amount from the customer's balance. Creating a record of the transaction to ensure the bank's database is updated with the latest transaction data. The transaction details are recorded in the system to maintain a trackable history.

6. Transaction Information: After updating the account, the system generates transaction information, which includes the details of the withdrawal (e.g., amount, date, time, location, etc.). This transaction information is then stored securely for future reference and reporting.

7. Database Update: The database is updated with the latest account and transaction data. This ensures the customer's account reflects the changes made during the withdrawal process.

The transaction is stored in the database for further auditing, reporting, and account management. By this method, customers can withdraw money either through a bank using

a cheque or via an ATM using account information. This process ensures that the customer's transaction is valid and that funds are available. After verification, the customer's account information is updated in the bank's database, reflecting the withdrawal. A secure record of the transaction is generated and stored for future use.

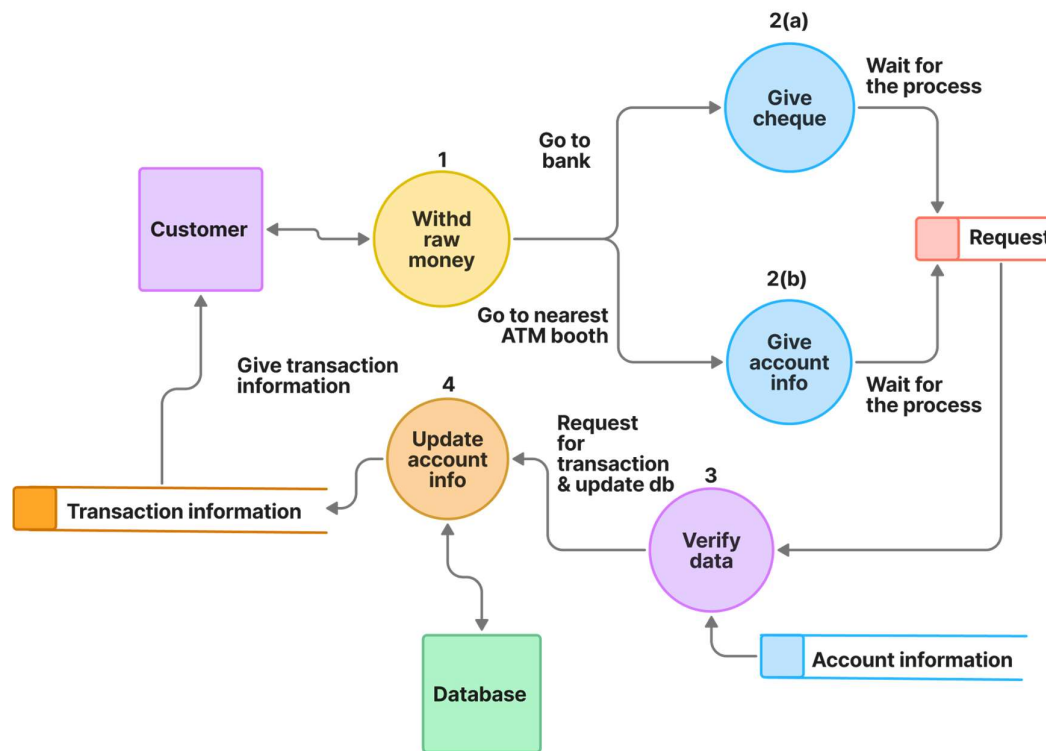


Figure 3.30: DFD For Money Withdraw

3.5 Detailed Feasibility Analysis

A feasibility study tests the viability of an idea, a project or even a new business. The goal of a feasibility study is to emphasize potential problems that could occur if a project should be pursued.

The purpose of the feasibility study is to present the project parameters and potential solutions to the defined problem. Project constraints and limitations of expenditure are among the various factors that will determine viability. After brainstorming various potential solutions, we tried to elaborate on each of the solutions. We have tried to obey three key considerations which are involved with feasibility analysis- Economic feasibility, Technical feasibility and Behavioral feasibility.

As we have studied the present system of IFIC Bank PLC, we have concluded three potential candidate systems, which are discussed in detail in the following sections.

3.5.1 Candidate System-I(Core Banking System Upgrade)

This system works on core banking system upgrades. The details of the system are discussed below:

1. Upgrading a core banking system is a complex process that involves replacing or enhancing the main software and hardware that a bank uses to manage its daily operations. The core banking system typically handles essential functions like transaction processing, account management, loan servicing, and more. The bank first evaluates why an upgrade is necessary, such as to improve performance, integrate new digital services (e.g., mobile banking), comply with new regulations, or reduce operational risks.

2. This system requires significant hardware and software upgrades, including server infrastructure, database scaling, and cloud integration. Also, additional requirements include high-level technical expertise, but the bank's current IT staff can handle the upgrade with some vendor support.

3. The system aims to provide improved transaction speed, scalability for future growth, and enhanced integration with digital banking platforms. It would streamline back-office operations and improve customer experience. Although staff would require training on new system functionalities, as the interface is familiar, the learning curve is moderate.

4. The financial analysis of the given system is shown below:

Initial investment: 3 million BDT.

Annual maintenance: 500,000 BDT.

Training: 100,000 BDT.

Improved operational efficiency, reduced transaction times, and fewer errors could lead to cost savings of 1.5 million (approximately) annually.

Break-even: (Around) 2.2 years.

5. Some challenges have arisen regarding this candidate system, some of which are transitioning from legacy systems could cause operational downtime, integration with third-party apps could be complex, staff could be reluctant to adapt, core policy change may take unprecedented time etc.

3.5.2 Candidate System-II (Mobile Banking Application Improvement)

This system works on improving mobile/digital banking systems. The details of the systems are discussed below:

1. This system focuses on enhancing the existing mobile platform by adding features, improving performance, and providing a better user experience. The goal is to make the system more user-friendly, secure, and efficient to meet evolving customer needs. This includes better UI/UX design, better security measures, optimizing load time, developing offline functionality, introducing new features, expanding service areas etc.

2. The system requires the re-development of mobile applications for iOS and Android, including robust security measures, encryption, and real-time transaction capabilities. Additional requirements include outsourcing to a mobile app development company with fin-tech experience. The bank's IT team can manage backend integration.

3. This system aims to enable customers to conduct banking transactions on the go, apply for loans, and manage investments, improving customer convenience, more conveniently. Minimal staff training is required. Customers would need user-friendly guidance to adapt to the changes in the app.

4. The financial analysis of the given system is shown below:

Development and launch: 1.5 million BDT.

Maintenance and updates: 200,000 BDT annually. Marketing and customer onboarding: 250,000 BDT.

However, the benefits include an expected increase in customer base, additional revenue from transaction fees, and reduced branch traffic, with a projected revenue increase of 2 million annually. Break-even can be considered approximately at 1.25 years.

5. Some of the major challenges include security risks and ensuring compatibility with the core banking system, customers' and staff' reluctance to adapt etc.

3.5.3 Candidate System-III(Cloud-Based Customer Relationship Management (CRM) System)

This system aims to develop a cloud-based customer relationship management process. The details of this system are discussed below:

1. A cloud-based Customer Relationship Management (CRM) system aims to help busi-

nesses manage their interactions with customers, streamline sales, marketing, and customer service processes, and improve overall customer satisfaction. By leveraging the cloud, these systems provide real-time access to data, collaboration tools, and analytics, allowing businesses to operate more efficiently and make informed decisions.

2. The main goal is to store and organize all customer information (e.g., contact details, transaction history, preferences) in a centralized, cloud-based database that can be accessed from anywhere. This system can provide a single source of truth for customer information, ensuring consistency and accuracy across departments (sales, marketing, customer service).

3. The system requires migration of customer data to a cloud-based platform, integration with existing banking systems, and ensuring data security. Though implementation would need external consultants, the bank's IT team can handle routine operations.

4. This positively impacts customer relationship management by offering personalized services, tracking customer interactions, and automating marketing campaigns. Improves staff efficiency and customer retention. Although significant training is required for customer-facing staff to utilize CRM tools effectively.

5. The financial analysis of the given system is shown below:

Implementation: 2 million BDT.

Licensing and cloud fees: 400,000 BDT annually.

Training and support: 150,000 BDT.

Benefits include expected customer retention by 15%, an increase in cross-selling opportunities, and an additional 1.8 million in annual revenue. Break-even is calculated approximately at 1.5 years.

3.5.4 Comparative Analysis of Candidate Systems

A comparative analysis of the proposed candidate system is shown below:

The evaluation matrices are also shown below:

Criteria	System 1: Core Banking Upgrade	System 2: Mobile Banking App	System 3: CRM System
Initial Cost(BDT)	300 million	150 million	200 million
Annual Maintenance(BDT)	5,00,00,000	2,00,00,000	4,00,00,000
Training Costs(BDT)	1,00,00,000	Minimal	1,50,00,000
Projected Annual Revenue(BDT)	150 million	200 million	180 million
Break-Even Period	2.2 years	1.25 years	1.5 years
Operational Impact	Improved transaction speeds, scalability	High customer convenience	Enhanced customer experience
Risks	Downtime during migration	Security Risks	Data migration and compliance

Figure 3.31: Comparative analysis of proposed candidate systems

Evaluation Criteria	Weighting Factor	Candidate1 Rating	Candidate1 Score	Candidate2 Rating	Candidate2 Score	Candidate3 Rating	Candidate3 Score
User-Friendly	5	3	18	5	25	4	20
Employee-Friendly	4	3	18	5	25	4	20
Use of Technology	4	4	20	5	25	5	25
Reliability	4	4	20	4	20	5	25

Figure 3.32: Performance Evaluation Matrix

Evaluation Criteria	Weighting Factor	Candidate1 Rating	Candidate1 Score	Candidate2 Rating	Candidate2 Score	Candidate3 Rating	Candidate3 Score
System Development	4	3	18	5	25	4	20
System operation	5	3	18	4	20	4	20
Technical Support	4	4	20	5	25	5	25

Figure 3.33: Cost Evaluation Matrix

Evaluation Criteria	Candidate System-1	Candidate System-2	Candidate System-3
Performance			
User-friendly	90%	98%	98%
Employee-friendly	90%	98%	95%
Use of Technology	95%	98%	98%
Reliability	95%	92%	90%
Cost			
System Development	High	Medium	Medium
System Operation	High	Medium	High
Technical Support	Medium	Low	Medium

Figure 3.34: Qualitative Evaluation Matrix

Upon analysis, it can be seen that, among all the proposed systems, candidate system-2 is the most efficient, both cost and time-wise.

3.5.5 Recommendation

Based on the feasibility analysis, System 2: Mobile Banking Application Improvement is the most feasible option in the short term due to its: Lower initial cost and shorter break-even period (1.25 years).

Significant customer demand for mobile banking solutions, which is likely to improve customer retention and attract new customers quickly. Lower technical risks, as it can be integrated with the bank's current systems with minimal downtime. Upgrading an existing mobile application typically requires a lower investment than overhauling the entire core banking system or launching a completely new platform. The bank can build on its existing mobile infrastructure rather than starting from scratch, reducing both time and costs. System 2 improvements can often be rolled out incrementally and more quickly compared to a complete system transformation. The bank can deploy new features, security upgrades,

and performance improvements in stages without disrupting daily operations. With features like a customizable dashboard and personalized recommendations, System 2 allows the bank to tailor the app to individual user needs. This results in a more engaging experience, leading to increased usage and customer satisfaction. Also, cybersecurity threats are a major concern in mobile banking, with phishing, fraud, and data breaches becoming more sophisticated a key aspect of System 2 is to enhance security through features like biometric authentication, two-factor authentication (2FA), and real-time fraud detection. This will protect both the bank and its customers from potential security threats, fostering trust in the system.

However, in the medium to long term, combining System 2 with System 3 (CRM) would provide a comprehensive solution by improving both customer-facing services and internal efficiency.

3.6 Cost-benefit Analysis

We know, cost and benefits analysis is a process by which organizations can estimate the cost and benefits of decisions, systems or projects to find the most effective alternatives. By providing a clear view of the results of an action, cost-benefit analysis is an extremely useful tool in developing business strategy, evaluating a new hire, or making resource allocation or purchase decisions. So, we developed a cost and benefit analysis model in this report. Our model is developed by defining the benefits of a decision as well as the associated costs and subtracting the costs from the benefits.

3.6.1 Cost Breakdown

A break-down of the costs involving the candidate system has been shown below:

Our goal is to improve the bank's online and mobile banking system to enhance customer experience, reduce manual processes, and attract new users. The total costs primarily involve software development, third-party services, hardware upgrades, licensing, staff training, annual IT support, marketing etc. The most costly part is developing the software for banking. With the annual ongoing cost being 2,350,000 BDT, the total cost of implementing a new software system would be 650,000 BDT.

3.6.2 Benefits Breakdown

A break-down of the costs involving the candidate system has been shown below:

As per our calculations, the main benefits of the candidate system include annual increased revenue, reduced transaction processing cost, reduced physical branch overhead and cost

Costs	Amount(BDT)
Software Development	15,00,00,000
Third-party Services (Annual)	2,00,00,000
Hardware Upgrades	4,00,00,000
Licensing and maintenance(Annual)	1,50,00,000
Staff Training	1,00,00,000
IT Support(Annual)	3,00,00,000
Marketing Campaign	2,50,00,000
Customer Incentives	1,00,00,000
Total Initial Costs:	23,50,00,000
Ongoing Annual Costs:	6,50,00,000

Figure 3.35: Cost Breakdown

Benefits	Amount(BDT)
Increase Revenue(Annual)	7,50,00,000
Reduced Transaction Processing Cost(Annual)	2,00,00,000
Reduced Physical Branch Overhead	2,50,00,000
Cost Saved By Improved Retention Rate	2,00,00,000
Total Annual Benefits:	14,00,00,000

Figure 3.36: Benefit Breakdown

saved by improved retention rate. The total benefits add up to 14,00,00,000 BDT, which surpasses the annual cost fairly.

3.6.3 Break-Even Analysis

A Break-Even Analysis helps determine when the benefits of an investment will cover its costs. For the mobile/online banking system improvement, we want to calculate the break-even point, where the total cumulative benefits equal the total cumulative costs. The summary of our cost-benefit analysis is shown below:

Initial Costs: This is the up-front investment needed to implement the system. In this case, it's 23,50,00,000 BDT.

Ongoing Costs: These are the yearly costs required to maintain the system (e.g., IT support, maintenance), which total 6,50,00,000 BDT annually.

Annual Benefits: The system is expected to generate 14,00,00,000 BDT in annual benefits (new revenue, cost savings, customer retention, etc.).

Net Annual Benefit: Each year, the system generates 7,50,00,000 BDT in net annual benefits (total annual benefits minus ongoing costs).

The break-even analysis table showing the cumulative costs and benefits over time, along with the net benefits each year, is given below:

Year	Cumulative Costs(BDT)	Cumulative Benefits(BDT)	Net Benefits(BDT)
o(Initial)	23,50,00,000	0	-23,50,00,000
1	30,00,00,000	14,00,00,000	-16,00,00,000
2	36,50,00,000	28,00,00,000	-8,50,00,000
3	43,00,00,000	42,00,00,000	-1,00,00,000
4	49,50,00,000	56,00,00,000	+6,50,00,000
5	56,00,00,000	70,00,00,000	+14,00,00,000

Figure 3.37: Cumulative Costs and Benefits Analysis

To calculate the break-even point in years, we use the formula where we divide initial costs(23,50,00,000 BDT) by net annual benefits (7,50,00,000 BDT). The result nearly sums up to 3.13 years. The break-even analysis graph is shown below:

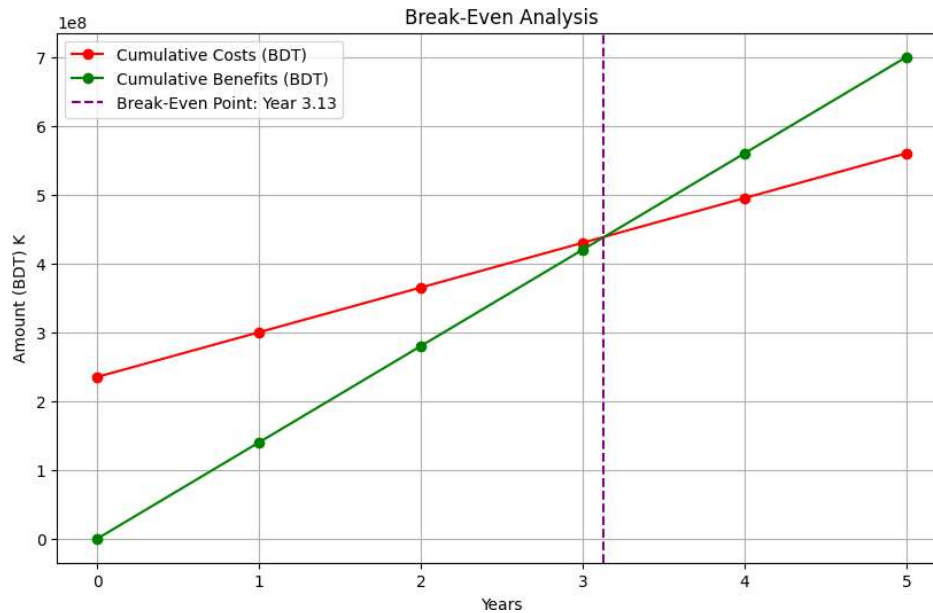


Figure 3.38: Break-Even Analysis Graph

3.7 Conclusion

A feasibility study is conducted to select the best system that meets performance requirements. This process includes the evaluation of a candidate system and the selection of the best system for the job. Three key considerations are mainly involved in the analysis- economic, technical and behavioural. We have tried to discuss in detail the existing system, identify its shortcomings and provide some suitable candidate systems. In conclusion, we have tried to present an entire overview of our detailed survey at IFIC Bank PLC.

Chapter 4

Design

4.1 Introduction

Both forms and databases are foundational to the success of any system. Forms are essential for efficient data entry and user interaction, while databases ensure the secure, scalable, and organized storage of that data. Together, they enable smooth operations, accurate data management, and informed decision-making, making them vital for any well-functioning system.

4.2 DFD of the Proposed Candidate System

The updated data flow diagram introduces a significant change aimed at strengthening the security of the customer interaction process, specifically focusing on preventing unauthorized access to sensitive account information. This change is represented by the red box at the top of the diagram, which describes the addition of an authentication step.

4.2.1 Inquiring Account Status

New Authentication Step (PIN/Password)

The introduction of an authentication phase ensures that customers must authenticate themselves using a PIN or password before proceeding with any transaction. This step adds an extra layer of security, protecting the system from unauthorized access to account information. By incorporating this measure, the system can verify the customer's identity at the beginning of the process, ensuring only legitimate users can access their account data and conduct transactions.

This addition enhances the **security posture** of the system without significantly altering the overall workflow. The authentication step acts as a safeguard, ensuring that the transaction

data and account details remain secure throughout the entire process.

Changes Due to Authentication Step:

Improved Security: By verifying user identity early in the process, the system ensures that sensitive information is accessible only to legitimate users.

Data Integrity: Ensures the data retrieved, transferred, or modified is handled in a secure manner and associated with an authenticated individual.

User Trust: Customers are more likely to trust a system that requires authentication before providing access to financial details.

Protects Against Data Breaches: Authentication protects both the user and the system from malicious actors who might try to gain unauthorized access.

The authentication step in the flowchart is a crucial addition to secure the customer's account data and transactional processes. It can be implemented using various methods (SFA, 2FA, MFA) depending on the desired security level. Ensuring strong authentication is essential to protecting both user data and the system from unauthorized access and potential fraud.

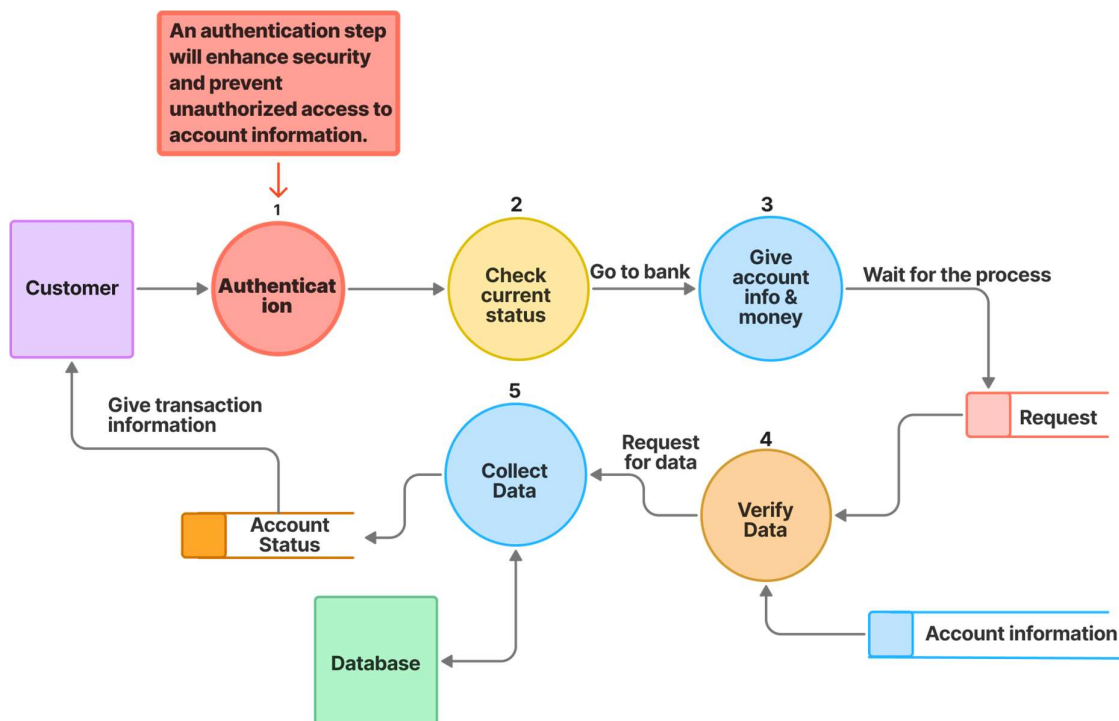


Figure 4.1: Proposed DFD of Knowing Account Status

4.2.2 Money Deposit

This updated diagram introduces two key improvements to the security and robustness of the system. In addition to the previously described authentication step, the new flow incorporates an encryption/decryption process for transaction information, ensuring that sensitive data is securely handled within the system. These updates are described in the two red boxes.

Authentication Step (PIN/Password)

As in the previous version, the system begins with an authentication phase where the customer must enter their PIN or password. This step is critical for securing access to the system, ensuring that only authorized users can initiate transactions or update account information.

Encryption/Decryption

The process involves encrypting transaction data before it is stored in the database and decrypting it when needed. This ensures that sensitive transaction data is stored securely, protecting it from unauthorized access or breaches.

Encryption of Transaction Information: Before transaction information is saved in the database, it is encrypted to make sure the data is unreadable and secure. Encryption adds a layer of security to sensitive data like account balances, transaction histories, or personal financial details.

The decryption of Transaction Information: When the data is needed (e.g., for user queries or financial analysis), it is decrypted to its original, readable form. This ensures that only authorized systems or individuals with decryption keys can access the information.

Changes Due to Encryption/Decryption Step:

Increased Security: Storing encrypted transaction data in the database makes it much harder for attackers to access sensitive financial information even if they manage to breach the database.

Compliance: Many regulations (such as GDPR or PCI-DSS in the financial sector) require encryption of sensitive data to protect customer information.

Cashflow Analysis: As highlighted, storing transaction information helps with investigating transaction history and analyzing user cash flow. With encryption in place, this analysis is secure and trustworthy.

The encryption/decryption step adds a robust layer of security for handling sensitive transaction data, preventing unauthorized access, ensuring regulatory compliance, and maintaining data integrity. It makes the overall system more secure and reliable when dealing with personal financial information.

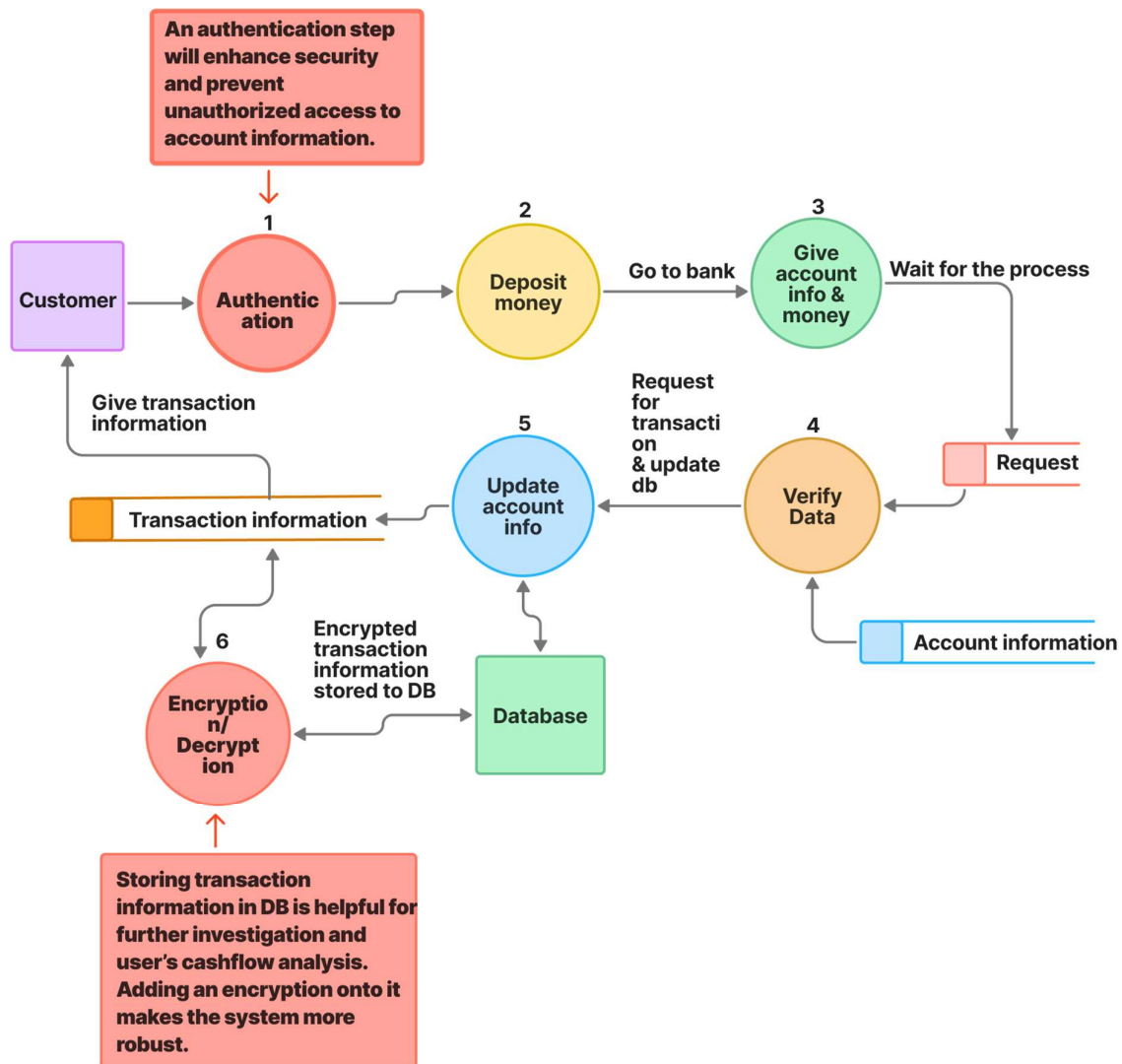


Figure 4.2: Proposed DFD of Money deposit

4.2.3 Money Withdrawal

Just like the Money Deposit process, the Money Withdrawal process adds Authentication and Encryption/Decryption steps as an enhancement. In the diagram, the changes highlighted with red boxes are:

Authentication Step: An additional authentication step (PIN/Password) is introduced after the customer initiates a transaction. This step is labelled as a measure to enhance security and prevent unauthorized access to account information. Authentication ensures that only authorized users can access and initiate sensitive transactions, such as withdrawing money. It verifies the identity of the customer before allowing them to interact with their account.

Encryption/Decryption Process: A new step at the end involves encryption/decryption of transaction information stored in the database. This step ensures the system's robustness by encrypting sensitive transaction data, which can be used for future investigation and cash flow analysis. These changes focus on strengthening the security aspects of the system by adding an extra layer of authentication and encrypting stored data. **Changes Due to the Proposed Steps**

Prevents Unauthorized Access: By requiring authentication (e.g., passwords, PINs, biometric scans), the system ensures that only the rightful account holder can withdraw money. This helps prevent unauthorized withdrawals, fraud, or identity theft.

User Verification at Different Touchpoints: Whether the withdrawal is made via cheque at a bank or through an ATM, authentication confirms the identity of the customer at the point of interaction. This prevents malicious actors from accessing the account even if they physically have access to the ATM or a cheque.

Enhanced Security During Account Interactions: Authentication protects all touchpoints (bank, ATM, or online) by providing a layer of security that ensures the user's identity is verified before money is withdrawn.

Multi-Factor Authentication (MFA): For even greater security, implementing multi-factor authentication (requiring multiple credentials like a combination of PINs, biometrics, and one-time codes) provides additional barriers for malicious actors attempting to impersonate a legitimate customer.

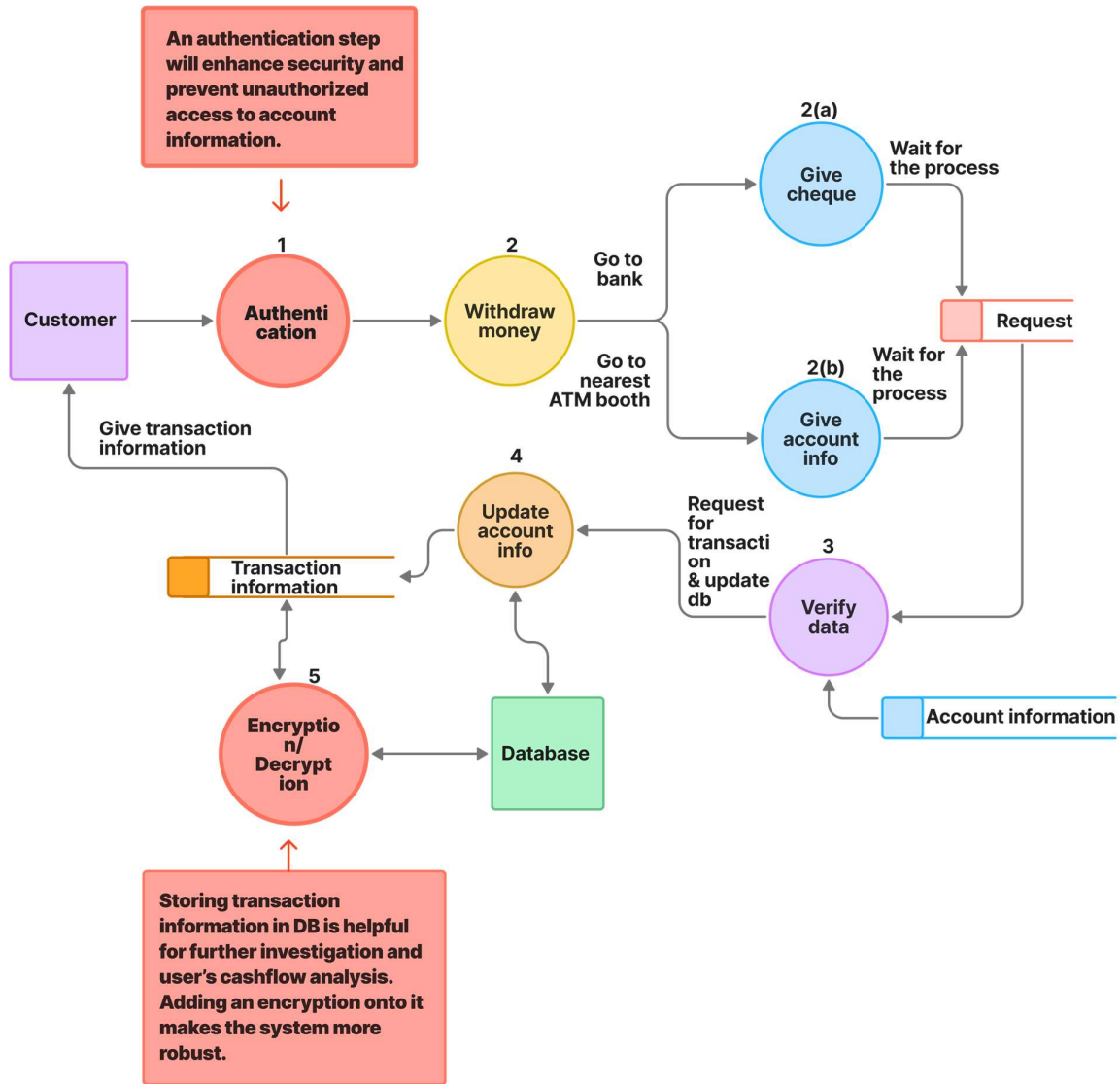


Figure 4.3: Proposed DFD of Money Withdraw

4.3 Database Design

A database is the system's core for storing, organizing, and managing data, ensuring that the information collected via forms is efficiently processed, retrieved, and analyzed. A well-designed database ensures that data is logically organized, typically in tables with relationships between them (e.g., customer data, transaction data). This structure enables the system to efficiently store and retrieve data, supporting smooth operations and allowing for complex queries and reports. Database design focuses on maintaining data integrity (accuracy and consistency) through the use of constraints, normalization, and relationships between entities (e.g., primary keys, and foreign keys). This ensures that data remains consistent, accurate, and up-to-date, avoiding duplication and conflicting information that could compromise business operations.

4.3.1 Database Schema

The skeleton structure of a database schema is the logical picture of the entire database. It outlines the structure of the data and the relationships that connect them. A database schema identifies the relationships between its entities. It includes a description of the database that can be illustrated using a schema diagram. The database designers create the schema to make the database easier for programmers to understand and use. In a banking system, the entities will typically involve customers, accounts, transactions, loans, and employees. Below is the proposed schema with entities and relationships:

Database Schema for Banking System:

- *Customer* (customer_id: int, name: string, address: string, nid: int, phone: int)
- *Account* (account_id: int, account_type: string, balance: decimal, date_opened: date, customer_id: int)
- *Transaction* (transaction_id: int, transaction_date: date, amount: decimal, transaction_type: string, account_id: int)
- *Loan* (loan_id: int, loan_type: string, loan_amount: decimal, interest_rate: decimal, loan_date: date, customer_id: int)
- *Branch* (branch_id: int, branch_name: string, address: string)
- *Employee* (employee_id: int, name: string, position: string, branch_id: int)
- *Loan_Payment* (payment_id: int, payment_date: date, amount: decimal, loan_id: int)
- *Account_Holder* (Junction Table for Customers with Multiple Accounts: customer_id: int, account_id: int, PRIMARY KEY (customer_id, account_id))

4.3.2 ER Diagram

An ER diagram (Entity-Relationship diagram) is a graphical representation of the entities (objects, concepts) and the relationships between them within a system, typically used in database design. ER diagrams help visualize how data is structured and how different elements within a system are interconnected. The components of an ER diagram include entities, attributes, relationships, cardinality, primary keys and foreign keys. ER diagrams provide a clear representation of the data structure and relationships, ensuring the logical organization of information. Helps database designers understand how data will be stored and managed, preventing data redundancy and ensuring data integrity.

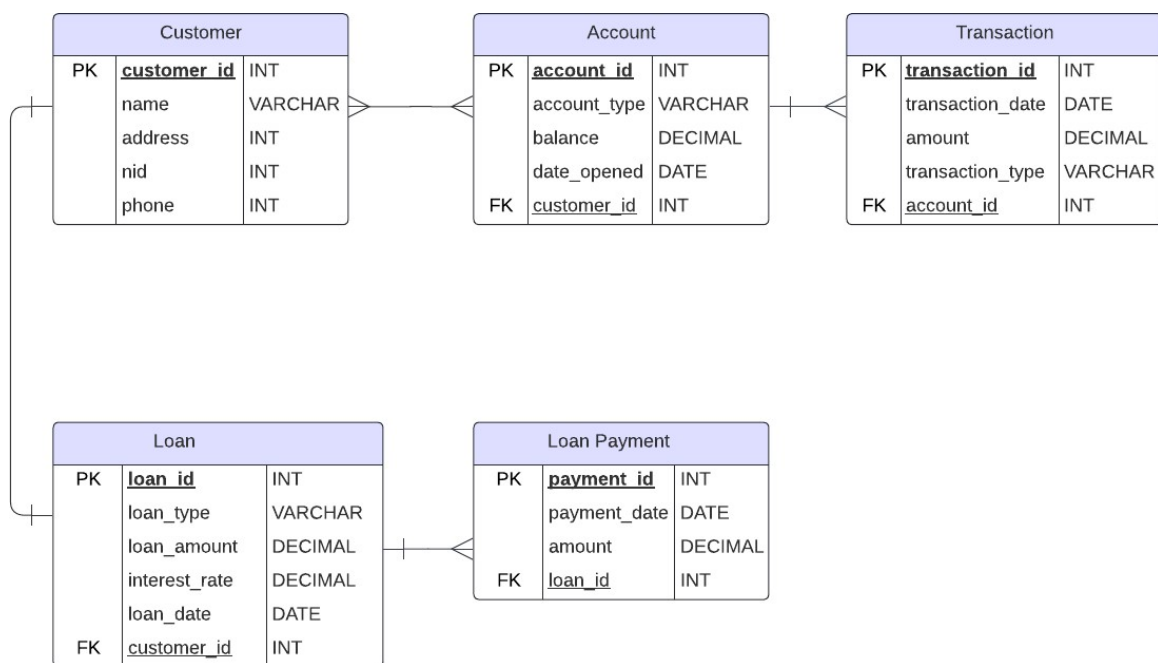


Figure 4.4: ER Diagram of the candidate system

4.4 Form Design

Forms act as the interface through which users interact with the system. They are critical for data collection, processing, and user engagement. Forms are designed to capture data in a structured and consistent manner, ensuring that users provide accurate and relevant information. Benefit: By creating well-designed forms, organizations can minimize user errors and ensure the quality of the data collected.

4.4.1 Details of findings:

The main form of IFIC Bank is the account opening form. The form is discussed below below:

The first part of the form contains fields like date, customer ID, party ID, account number etc. which are to be filled out by the bank.

The upper right corner is reserved for attaching the customer's passport-sized photo.

The second part contains fields for filling out account-related information and the third part contains a file for filling out the personal information of the customer.

The form consists of two pages, where the first page contains questions about the customer's information and the second page contains questions about the nominee's information and space for declaration and signature.

The overall design of the form is detailed and comprehensive, although the design seems a bit compact and messy.

The form needs to be redesigned to improve the visuals and be easier for the customers to read and fill out.

4.4.2 Recommendations

Some of our recommendations based on analysis of the forms are given below:

Separating the parts belonging to banks and customers: As per the original design, the part that is reserved for filling out by the bank is almost segregated from the other parts. Also the text "For bank use only" is written in a very small font and is hard to read. This could confuse new customers.

Using larger font size: The font used in the form is very small. Even if it helps to limit the size of the form to two pages, it makes the form difficult to read. Using a larger font size could help solve this problem.

Using a more sophisticated blend of colours: The form could use better colour coordination. Some of the writings are done on a green background, which makes it harder to read.

Redesigning the nominee's section: The nominee's section seems more compact and messy than the rest of the form. There is barely enough space to fill up the information. Also, the instructions are written in very small spaces which makes them almost unreadable. This part alone hampers the overall aesthetic of the form. This part needs to be rearranged

neatly.

Increasing the space reserved for signature: The space reserved for signature is comparatively smaller than other fields in the form. Since, the customer has to write his name, and mobile number and then sign the document, that field must have enough space.

4.5 UI Design

Since the IFIC Bank Mobile app has several problems per the users for example: the "Forgot Password" option does not work, the absence of auto-fill options etc, we focused on an interactive user-friendly design of UI. A mobile app nowadays works as a digital form. To improve the user interface (UI) and functionality of the IFIC Bank mobile app and online platform, we focused on the following key areas:

1. User-Friendly Design: The new design has a streamlined navigation system, simplified menus and options to make navigation intuitive. It uses clear labels, icons, and visual cues. It also has a modern aesthetic. It updates the UI with a cleaner, more modern look, using a consistent colour scheme, typography, and spacing.

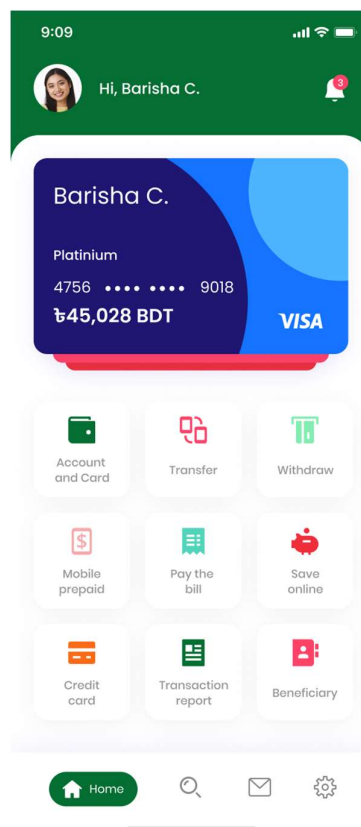


Figure 4.5: Home Page of the New App

2. Save Transaction Data: Transaction Autofill: The new design implements a system that saves frequently used data (like account numbers, names, and branches) securely to minimize user input for repeated transactions.

History & Recurrence: It allows users to easily repeat past transactions by showing a history of previous ones and enabling recurring payments or transfers.

Figure 4.6: Transfer Money

Figure 4.7: Transfer Money

Figure 4.8: Autofill of Sensitive Information

3. Credential Recovery: The new design fixed the "Forgot Password" feature. It ensures the "Forgot Password" function works reliably. It also provides multiple recovery methods (email, SMS, security questions, or biometric authentication).

It also provides Multi-Factor Authentication(MFA) and offers users the ability to add an extra layer of security for login through OTPs (one-time passwords) or biometrics like fingerprints or face IDs.

9:09

< Forgot password

Type your phone number

(+880)

We texted you a code to verify your phone number

Send

1 2 3

4 5 6

7 8 9

. 0

Figure 4.9: Forgot Password

9:09

< Forgot password

Type your phone number

(+880) 175019 2345

We texted you a code to verify your phone number

Send

1 2 3

4 5 6

7 8 9

. 0

Figure 4.10: Forgot Password

Figure 4.11: Credential Recovery

4.6 Conclusion

Both forms and databases are foundational to the success of any system. Forms are essential for efficient data entry and user interaction, while databases ensure the secure, scalable, and organized storage of that data. Together, they enable smooth operations, accurate data management, and informed decision-making, making them vital for any well-functioning system.